Notes on Binary Outcome Model

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I synthesized the theories of the binary outcome model in this study note. Some topics rarely covered in Econometrics textbooks e.g., separation, generalized linear model, Cox and Snell's R^2, HL test, etc. are also mentioned. This note also covers the treatment of the binary outcome in the framework of Causal Inference (LPM in Section 1.13) and Structural Econometrics (Single Agent Dynamic Discrete Choice Model in Section 2.6).

1 General Binary Outcome Model

$$y_i = \begin{cases} 1 & \text{with probability } p_i \\ 0 & \text{with probability } 1 - p_i \end{cases}$$

Model p_i as

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = F(\boldsymbol{x}_i'\boldsymbol{\beta}) = F_i$$

1.1 Marginal Effect

$$\begin{aligned}
\frac{\partial Pr(y_i = 1|\boldsymbol{x}_i)}{\partial \boldsymbol{x}_i} &= \frac{\partial F(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{x}_i} \\
&= \frac{\partial F(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{x}_i'\boldsymbol{\beta}} \frac{\partial \boldsymbol{x}_i'\boldsymbol{\beta}}{\partial \boldsymbol{x}_i} \\
&= F'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{\beta} && \text{If } F(.) \text{ is cdf, } F'(.) > 0. \text{ So, } sign(\boldsymbol{\beta}) \text{ decide the sign}
\end{aligned}$$

Average Marginal Effect is

$$AME := N^{-1} \sum_{i=1}^{N} F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\widehat{\boldsymbol{\beta}}$$

or

$$AME := F'(N^{-1} \sum_{i=1}^{N} \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\widehat{\boldsymbol{\beta}} = F'(\overline{\boldsymbol{x}}'\widehat{\boldsymbol{\beta}})\widehat{\boldsymbol{\beta}}$$

If $\overline{\boldsymbol{x}}$ includes $\overline{ln(z)}$, it can be replaced by $ln(\overline{z})$. If $\boldsymbol{x}_i$ includes z_i and z_i^2, AME_i for z_i is $F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(\widehat{\beta}_z + 2\widehat{\beta}_z z_i)$. Therefore, AME is $F'(\overline{\boldsymbol{x}}'\widehat{\boldsymbol{\beta}})(\widehat{\beta}_z + 2\widehat{\beta}_z \overline{z})$ where $\overline{z^2}$ in $\overline{\boldsymbol{x}}$ can be replaced by $\overline{z}^2$. If $\boldsymbol{x}_i$ includes an indicator z_i, AME is $F(\overline{\boldsymbol{x}}'_{-z}\widehat{\boldsymbol{\beta}}_{-z} + 1 \cdot \widehat{\beta}_z) - F(\overline{\boldsymbol{x}}'_{-z}\widehat{\boldsymbol{\beta}}_{-z})$.

1.2 Maximum Likelihood Estimation

1.2.1 Probability Mass Function

As y_i is binary, it must be Bernoulli distributed. The probability mass function (pmf) of such random variable is

$$\begin{aligned}
f(y_i|\boldsymbol{x}_i) &= Pr(y_i = 1|\boldsymbol{x}_i)^{y_i} Pr(y_i = 0|\boldsymbol{x}_i)^{1-y_i} \\
&= Pr(y_i = 1|\boldsymbol{x}_i)^{y_i} (1 - Pr(y_i = 1|\boldsymbol{x}_i))^{1-y_i} \\
&= p_i^{y_i}(1-p_i)^{1-y_i} \\
&= F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1 - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}
\end{aligned}$$

1.2.2 Log Likelihood Function

$$\begin{aligned}
ln[L_N(\boldsymbol{\beta})] &= ln[\prod_{i=1}^{N} f(y_i|\boldsymbol{x}_i)] && \text{assume independence} \\
&= ln[\prod_{i=1}^{N} F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1 - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}] \\
&= \sum_{i=1}^{N} ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1 - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}] \\
&= \sum_{i=1}^{N} \{y_i ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})] + (1-y_i) ln[1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})]\}
\end{aligned}$$

1.2.3 Gradient Vector

$$\begin{aligned}
\frac{\partial ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} &= \frac{\partial \sum_{i=1}^{N}\{y_i ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})] + (1-y_i) ln[1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})]\}}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^{N} \frac{\partial \{y_i ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})] + (1-y_i) ln[1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})]\}}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^{N} \{\frac{\partial y_i ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} + \frac{\partial (1-y_i) ln[1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}}\} \\
&= \sum_{i=1}^{N} \{y_i \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta})} F'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{x}_i + (1-y_i)\frac{1}{1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})}(-1)F'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{x}_i\} \\
&= \sum_{i=1}^{N} \{\frac{y_i}{F(\boldsymbol{x}_i'\boldsymbol{\beta})} F'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{x}_i - \frac{1-y_i}{1 - F(\boldsymbol{x}_i'\boldsymbol{\beta})} F'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{x}_i\} \\
&= \sum_{i=1}^{N} \{\frac{y_i}{F_i} F_i'\boldsymbol{x}_i - \frac{1-y_i}{1-F_i} F_i'\boldsymbol{x}_i\} \\
&= \sum_{i=1}^{N} \frac{y_i F_i'\boldsymbol{x}_i(1-F_i) - (1-y_i)F_i'\boldsymbol{x}_i F_i}{F_i(1-F_i)} \\
&= \sum_{i=1}^{N} \frac{(y_i F_i'\boldsymbol{x}_i - y_i F_i'\boldsymbol{x}_i F_i) - (F_i'\boldsymbol{x}_i F_i - y_i F_i'\boldsymbol{x}_i F_i)}{F_i(1-F_i)} \\
&= \sum_{i=1}^{N} \frac{y_i F_i'\boldsymbol{x}_i - F_i'\boldsymbol{x}_i F_i}{F_i(1-F_i)} \\
&= \sum_{i=1}^{N} \frac{y_i - F_i}{F_i(1-F_i)} F_i'\boldsymbol{x}_i
\end{aligned}$$

1.2.4 First Order Condition

$$\sum_{i=1}^{N} \frac{y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))} F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})\boldsymbol{x}_i = \mathbf{0}$$

There is no closed form solution of $\widehat{\boldsymbol{\beta}}_{mle}$ for the common choice of $F(.)$. We usually solve it with Newton's method or Fisher's scoring (see Appendix A and B). If the dataset is large, it is usually solved by stochastic gradient descent. If $F(.)$ is logistic function or standard normal cdf, $ln[L_N(\boldsymbol{\beta})]$ is strictly concave because its Hessian matrix is negative definite (see Appendix D). On the convex domain, the first order condition becomes a sufficient condition (Theorem 7.15 of Sundaram (1996)) and there is a unique maximizer (Theorem 7.14 of Sundaram (1996)) if it exists (note that the maximizer does not exist for logistic $F(.)$ when there is complete separation, see Section 6). For log-concave $F(.)$, the Hessian matrix is negative semi-definite (see Appendix D) and thus $ln[L_N(\boldsymbol{\beta})]$ is concave. If $F(-z) = 1 - F(z)$, the first order condition can be written as $\sum_{i=1}^{N}(2y_i-1)\frac{1}{F((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}F'((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})\boldsymbol{x}_i = \mathbf{0}$

1.2.5 Consistency of Maximum Likelihood Estimator

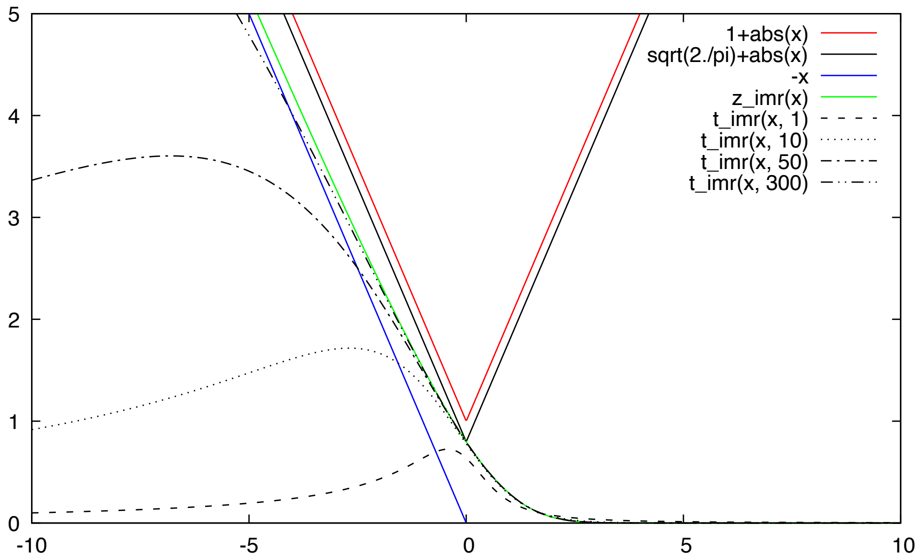
Theorem 1 (Theorem 2.5 of Newey and McFadden (1994))
Suppose that $z_i = (y_i, \boldsymbol{x}_i')'$ are i.i.d. with probability density or mass function $f(z_i;\boldsymbol{\theta}_0)$ and (i) $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0 \implies f(z_i;\boldsymbol{\theta}) \neq f(z_i;\boldsymbol{\theta}_0)$. (ii) $\boldsymbol{\theta}_0 \in \Theta$ which is compact. (iii) $lnf(z_i;\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta} \in \Theta$ with probability one. (iv) $\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta}|lnf(z_i;\boldsymbol{\theta})|] < \infty$. Then $\widehat{\boldsymbol{\theta}}_{mle} \to_p \boldsymbol{\theta}_0$.

Lemma 1
If $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is positive definite, then (i) holds if $F(.)$ is strictly monotonic. (iv) holds if the inverse Mills ratio $\frac{f(z)}{F(z)} < 1 + |z|$ where $f = F'$.

Proof: For $\boldsymbol{\beta} \in \Theta$ and $\boldsymbol{\beta} \neq \boldsymbol{\beta}_0$, the positive definiteness of $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ implies that $(\boldsymbol{\beta}-\boldsymbol{\beta}_0)'\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'](\boldsymbol{\beta}-\boldsymbol{\beta}_0) > 0 \iff \mathbb{E}[(\boldsymbol{x}_i'(\boldsymbol{\beta}-\boldsymbol{\beta}_0))^2] > 0 \iff (\boldsymbol{x}_i'(\boldsymbol{\beta}-\boldsymbol{\beta}_0))^2 > 0 \iff \boldsymbol{x}_i'(\boldsymbol{\beta}-\boldsymbol{\beta}_0) \neq 0 \iff \boldsymbol{x}_i'\boldsymbol{\beta} \neq \boldsymbol{x}_i'\boldsymbol{\beta}_0$. As $F(.)$ is strictly monotonic, $F(\boldsymbol{x}_i'\boldsymbol{\beta}) \neq F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$ and thus $f(y_i|\boldsymbol{x}_i;\boldsymbol{\beta}) := F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i} \neq F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i} =: f(y_i|\boldsymbol{x}_i;\boldsymbol{\beta}_0)$. Therefore, (i) holds. If $F(.)$ is a standard normal cdf or a standard logistic cdf i.e., a logistic function, the inverse Mills ratio $\frac{f(z)}{F(z)}$ is $\frac{\phi(z)}{\Phi(z)}$ and $\frac{1}{1+e^z}$, respectively. It can be shown that for any z, $\frac{\phi(z)}{\Phi(z)} \leq \sqrt{2/\pi} + |z| < 1 + |z|$ and $\frac{1}{1+e^z} = \Lambda(-z) = 1 - \Lambda(z) < 1 \leq 1 + |z|$. The inverse Mills ratio of the standard t random variable seems to be bounded by $1 + |z|$ numerically.

For any a ($\widetilde{a}$ is between 0 and a),

$$\begin{aligned}
|lnF(a)| &= |lnF(0) + \frac{f(\widetilde{a})}{F(\widetilde{a})}(a-0)| && \text{by Mean Value Theorem}\\
&\leq |lnF(0)| + \frac{f(\widetilde{a})}{F(\widetilde{a})}|a| && \text{Triangle inequality and } f(.) \text{ and } F(.) \text{ must be positive}\\
&< |lnF(0)| + (1+|a|)|a|\\
&= |lnF(0)| + |a| + |a|^2
\end{aligned}$$



$$
\begin{aligned}
|lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})| &= |y_i lnF(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1-y_i)lnF(-\boldsymbol{x}_i'\boldsymbol{\beta})| \\
&\leq y_i|lnF(\boldsymbol{x}_i'\boldsymbol{\beta})| + (1-y_i)|lnF(-\boldsymbol{x}_i'\boldsymbol{\beta})| && \text{by Triangle inequality} \\
&\leq |lnF(\boldsymbol{x}_i'\boldsymbol{\beta})| + |lnF(-\boldsymbol{x}_i'\boldsymbol{\beta})| && y_i \leq 1 \text{ and } 1-y_i \leq 1 \\
&< 2\cdot(|lnF(0)| + |\boldsymbol{x}_i'\boldsymbol{\beta}| + |\boldsymbol{x}_i'\boldsymbol{\beta}|^2) && \text{by inequality above} \\
&\leq 2\cdot(|lnF(0)| + ||\boldsymbol{x}_i||_2||\boldsymbol{\beta}||_2 + ||\boldsymbol{x}_i||_2^2||\boldsymbol{\beta}||_2^2) && \text{by Cauchy-Schwartz inequality} \\
&= 2\cdot(|lnF(0)| + \sqrt{\sum_j x_{ij}^2}||\boldsymbol{\beta}||_2 + \sum_j x_{ij}^2||\boldsymbol{\beta}||_2^2)
\end{aligned}
$$

Thus, $\mathbb{E}[|lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})|] < 2\cdot(|lnF(0)| + \mathbb{E}[\sqrt{\sum_j x_{ij}^2}]||\boldsymbol{\beta}||_2 + \sum_j \mathbb{E}[x_{ij}^2]||\boldsymbol{\beta}||_2^2)$. The positive definiteness of $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ implies $\mathbb{E}[x_{ij}^2] < \infty$ for $\forall j$. Thus, $\sum_j \mathbb{E}[x_{ij}^2] = \mathbb{E}[\sum_j x_{ij}^2] < \infty$, which implies $\mathbb{E}[\sqrt{\sum_j x_{ij}^2}] < \infty$. Thus, $\mathbb{E}[|lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})|] < \infty$ for any $\boldsymbol{\beta}$. This implies $\mathbb{E}[sup_{\boldsymbol{\beta}\in\Theta}|lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})|] < \infty$. Therefore, (iv) holds. Q.E.D.

Theorem 2 (Consistency of Probit and Logit MLE)
If $y_i|\boldsymbol{x}_i$ are i.i.d. with a probability mass function $F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i}$ where $F(.)$ is a standard normal cdf or a standard logistic cdf i.e., a logistic function, and $\boldsymbol{\beta}_0 \in \Theta$ which is compact and $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is a positive definite. Then, $\widehat{\boldsymbol{\beta}}_{mle} \to_p \boldsymbol{\beta}_0$.

Proof: By Lemma 1, (i) and (iv) of Theorem 1 holds. (ii) of Theorem 1 holds by hypothesis. $ln[F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}]$ is a continuous function of $\boldsymbol{\beta}\in\Theta$ i.e., (iii) of Theorem 1 holds. Q.E.D.

1.2.6 Asymptotic Normality of Maximum Likelihood Estimator

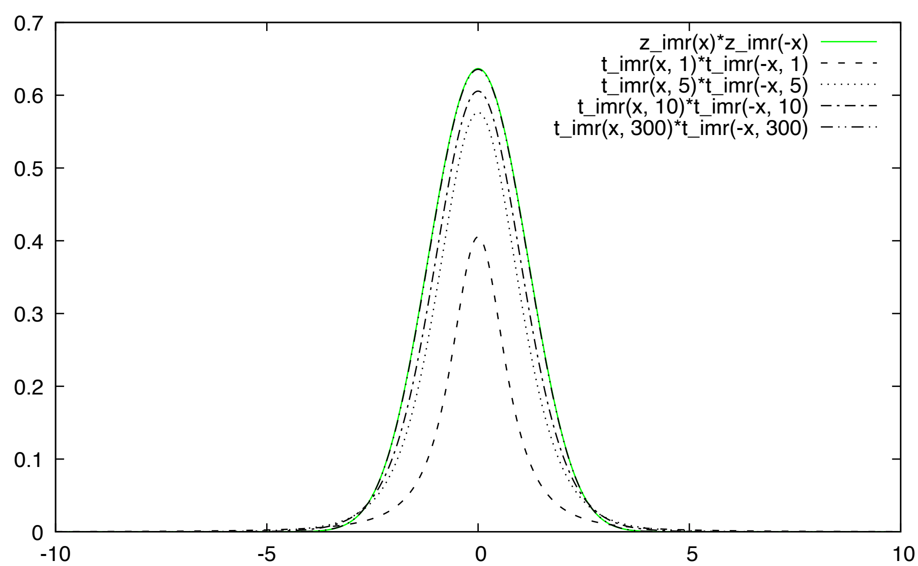
Theorem 3 (Theorem 3.3 of Newey and McFadden (1994))
Suppose that $z_i = (y_i, \boldsymbol{x}_i')'$ are i.i.d. and the assumptions of Theorem 1 are satisfied and (i) $\boldsymbol{\theta}_0 \in interior(\Theta)$. (ii) $f(z_i;\boldsymbol{\theta})$ is twice continuously differentiable with respect to $\boldsymbol{\theta}$ and $f(z_i;\boldsymbol{\theta}) > 0$ in a neighborhood N of $\boldsymbol{\theta}_0$. (iii) Sufficient conditions for interchanging integration and differentiation. (iv) $\boldsymbol{J} := \mathbb{E}[\nabla_{\boldsymbol{\theta}} lnf(z_i;\boldsymbol{\theta}_0)(\nabla_{\boldsymbol{\theta}} lnf(z_i;\boldsymbol{\theta}_0))']$ exists and is non-singular. (v) $\mathbb{E}[sup_{\boldsymbol{\theta}\in N}||\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}} lnf(z_i;\boldsymbol{\theta})||] < \infty$. Then $\sqrt{N}(\widehat{\boldsymbol{\theta}}_{mle} - \boldsymbol{\theta}_0) \to_d N(\mathbf{0}, \boldsymbol{J}^{-1})$.

Lemma 4
The positive definiteness of $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ implies (iv) and (v) if $F(.)$ is a standard normal cdf $\Phi(.)$ or a standard logistic cdf.

Proof:

$$
\begin{aligned}
\boldsymbol{J} &:= \mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}_0} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] \\
&= \mathbb{E}[\mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}_0} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i]] \\
&= \mathbb{E}[\mathbb{E}(\frac{y_i - F_i}{F_i(1-F_i)} F_i' \boldsymbol{x}_i \cdot \frac{y_i - F_i}{F_i(1-F_i)} F_i' \boldsymbol{x}_i'|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i)] \\
&= \mathbb{E}[\mathbb{E}(\frac{y_i - F_i}{F_i(1-F_i)} \frac{y_i - F_i}{F_i(1-F_i)} F_i'^2 \boldsymbol{x}_i \boldsymbol{x}_i'|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i)] \\
&= \mathbb{E}[\frac{\mathbb{E}[(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2|\boldsymbol{x}_i]}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2 \boldsymbol{x}_i \boldsymbol{x}_i'] \\
&= \mathbb{E}[\frac{\mathbb{E}[(y_i - Pr(y_i=1|\boldsymbol{x}_i))^2|\boldsymbol{x}_i]}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2 \boldsymbol{x}_i \boldsymbol{x}_i'] \qquad \text{if correctly specify CEF} \\
&= \mathbb{E}[\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2 \boldsymbol{x}_i \boldsymbol{x}_i']
\end{aligned}
$$

If $F(.) = \Phi(.)$, $\boldsymbol{J} = \mathbb{E}[\frac{\phi(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{\Phi(\boldsymbol{x}_i'\boldsymbol{\beta}_0)} \frac{\phi(-\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{\Phi(-\boldsymbol{x}_i'\boldsymbol{\beta}_0)} \boldsymbol{x}_i \boldsymbol{x}_i']$. It can be shown with L'Hopital's rule that $0 < \frac{\phi(z)}{\Phi(z)} \frac{\phi(-z)}{\Phi(-z)} < \infty$ for any z. $\frac{\phi(\infty)}{\Phi(\infty)} \frac{\phi(-\infty)}{\Phi(-\infty)} = \frac{0}{1}\frac{0}{0} = \frac{0}{0}$ and thus L'Hopital's rule can be applied. The derivative of $\phi(z)\phi(-z)$ is $\phi'(z)\phi(-z) - \phi(z)\phi'(-z) \to 0$ as $z \to \infty$. $lim_{z\to\infty} \frac{\phi(z)}{\Phi(z)} \frac{\phi(-z)}{\Phi(-z)} = lim_{z\to\infty} \frac{\phi'(z)\phi(-z) - \phi(z)\phi'(-z)}{\phi(z)\Phi(-z) - \Phi(z)\phi(-z)} = 0$. $\frac{\phi(0)}{\Phi(0)} \frac{\phi(-0)}{\Phi(-0)} = (\frac{1/\sqrt{2\pi}}{1/2})^2 = 2/\pi = 0.6366$. If $F(.)$ is a logistic function, $\boldsymbol{J} = \mathbb{E}[\frac{1}{1+e^{\boldsymbol{x}_i'\boldsymbol{\beta}_0}} \frac{1}{1+e^{-\boldsymbol{x}_i'\boldsymbol{\beta}_0}} \boldsymbol{x}_i \boldsymbol{x}_i'] = \mathbb{E}[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\Lambda(-\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i \boldsymbol{x}_i']$ and $0 < \Lambda(z)\Lambda(-z) = \Lambda(z)(1-\Lambda(z)) < 1$ for any z. By the positive definiteness of $\mathbb{E}[\boldsymbol{x}_i \boldsymbol{x}_i']$, $\boldsymbol{J}$ exists and is a positive definite (thus non-singular). Thus, (iv) holds. Q.E.D.



Theorem 5 (Asymptotic Normality of Probit and Logit MLE)
If $y_i|\boldsymbol{x}_i$ are i.i.d. with a probability mass function $F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i}$ where $F(.)$ is a standard normal cdf or a standard logistic cdf i.e., a logistic function, and $\boldsymbol{\beta}_0 \in interior(\Theta)$ and Θ is compact and $\mathbb{E}[\boldsymbol{x}_i \boldsymbol{x}_i']$ is a positive definite. Then,

$$
\sqrt{N}(\widehat{\boldsymbol{\beta}}_{mle} - \boldsymbol{\beta}_0) \to_d N(\boldsymbol{0}, \boldsymbol{J}^{-1}) = N(\boldsymbol{0}, \{\mathbb{E}[\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))} F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2 \boldsymbol{x}_i \boldsymbol{x}_i']\}^{-1})
$$

Proof: (i) of Theorem 3 holds by hypothesis. $F(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}$ is twice continuously differentiable and > 0 and thus (ii) of Theorem 3 holds. By Lemma 4, (iv) and (v) of Theorem 3 holds. Q.E.D.

Remark: if $\boldsymbol{x}_i$ is assumed to be non-stochastic/fixed, the traditional MLE theory depending on the i.i.d. assumption of y_i cannot be applied as the probability mass function of y_i depends on $\boldsymbol{x}_i$ and thus y_i is not identically distributed. This is the reason why Amemiya (1985, p.270-273) applies the theorems of the extremum estimator directly in which i.i.d are not assumed.

1.2.7 Moments

$\mathbb{E}(y_i|\boldsymbol{x}_i) = 1 \cdot Pr(y_i = 1|\boldsymbol{x}_i) + 0 \cdot Pr(y_i = 0|\boldsymbol{x}_i) = Pr(y_i = 1|\boldsymbol{x}_i) = F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$

$Var(y_i|\boldsymbol{x}_i) = \mathbb{E}(y_i^2|\boldsymbol{x}_i) - \mathbb{E}(y_i|\boldsymbol{x}_i)^2 = 1^2 \cdot Pr(y_i = 1|\boldsymbol{x}_i) - (1 \cdot Pr(y_i = 1|\boldsymbol{x}_i))^2 = Pr(y_i = 1|\boldsymbol{x}_i) - Pr(y_i = 1|\boldsymbol{x}_i)^2 = Pr(y_i = 1|\boldsymbol{x}_i)(1 - Pr(y_i = 1|\boldsymbol{x}_i))$

1.2.8 Information Matrix Equality

The direct derivation of the information matrix of the binary outcome model is clumsy because of the use of product rule.

$$\boldsymbol{I}(\boldsymbol{\beta}_0) := -\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] = \mathbb{E}[-\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i]] = \mathbb{E}[\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i']$$

$$\begin{aligned}
-\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i] &= -\mathbb{E}[\frac{\partial \frac{y_i}{F_i}F_i'\boldsymbol{x}_i - \frac{1-y_i}{1-F_i}F_i'\boldsymbol{x}_i}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i] = -\mathbb{E}[\frac{\partial(\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})F_i'\boldsymbol{x}_i}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i] \\
&= -\mathbb{E}[\frac{\partial(\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})}{\partial\boldsymbol{\beta}'}F_i'\boldsymbol{x}_i|_{\boldsymbol{\beta}_0} + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})\frac{\partial F_i'\boldsymbol{x}_i}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i] \\
&= -\mathbb{E}[(\frac{\partial\frac{y_i}{F_i}}{\partial\boldsymbol{\beta}'} - \frac{\partial\frac{1-y_i}{1-F_i}}{\partial\boldsymbol{\beta}'})F_i'\boldsymbol{x}_i + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})\frac{\partial F_i'\boldsymbol{x}_i}{\partial\boldsymbol{\beta}'}|\boldsymbol{x}_i]|_{\boldsymbol{\beta}_0} \\
&= -\mathbb{E}[(-\frac{y_i}{F_i^2}F_i'\boldsymbol{x}_i' - \frac{1-y_i}{(1-F_i)^2}F_i'\boldsymbol{x}_i')F_i'\boldsymbol{x}_i + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})F_i^{''}\boldsymbol{x}_i\boldsymbol{x}_i'|\boldsymbol{x}_i]|_{\boldsymbol{\beta}_0} \\
&= -\mathbb{E}[-(\frac{y_i}{F_i^2} + \frac{1-y_i}{(1-F_i)^2})F_i'^2\boldsymbol{x}_i\boldsymbol{x}_i' + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})F_i^{''}\boldsymbol{x}_i\boldsymbol{x}_i'|\boldsymbol{x}_i]|_{\boldsymbol{\beta}_0} \\
&= (\frac{\mathbb{E}[y_i|\boldsymbol{x}_i]}{F_i^2} + \frac{1-\mathbb{E}[y_i|\boldsymbol{x}_i]}{(1-F_i)^2})F_i'^2\boldsymbol{x}_i\boldsymbol{x}_i'|_{\boldsymbol{\beta}_0} - (\frac{\mathbb{E}[y_i|\boldsymbol{x}_i]}{F_i} - \frac{1-\mathbb{E}[y_i|\boldsymbol{x}_i]}{1-F_i})F_i^{''}\boldsymbol{x}_i\boldsymbol{x}_i'|_{\boldsymbol{\beta}_0} \\
&= (\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)} + \frac{1}{1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)})F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= \frac{(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)) + F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i'
\end{aligned}$$

For the binary outcome model, a correct specification of the CEF i.e., $Pr(y_i = 1|\boldsymbol{x}_i) = F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$ implies $\boldsymbol{B} = -\boldsymbol{A}$,

$$\begin{aligned}
\boldsymbol{B} &:= \mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}_0} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] \\
&= \mathbb{E}[\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i'] && \text{see the proof of Lemma 4 above} \\
&= -\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i)}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] := -\boldsymbol{A} = \boldsymbol{I}(\boldsymbol{\beta}_0)
\end{aligned}$$

Note that Amemiya (1985, p.272-273) assumes $\boldsymbol{x}_i$ to be non-stochastic and thus y_i is not identically distributed. The information matrix with the independence assumption is $\boldsymbol{I}(\boldsymbol{\beta}_0) := -lim_{N\to\infty}\frac{1}{N}\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] = -lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^N\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]$. If $y_i|\boldsymbol{x}_i$ is i.i.d., $\boldsymbol{I}(\boldsymbol{\beta}_0) := -lim_{N\to\infty}\frac{1}{N}N\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] = -\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]$. Note also that i.i.d. of $(y_i, \boldsymbol{x}_i')'$ does not necessarily imply i.i.d. of $y_i|\boldsymbol{x}_i$. E.g., $\mathbb{E}[y_i|\boldsymbol{x}_i] = g(\boldsymbol{x}_i'\boldsymbol{\beta})$ depends on $\boldsymbol{x}_i$ and thus $y_i|\boldsymbol{x}_i$ is not identically distributed.

1.2.9 Estimated Covariance Matrix

We follow Newey and McFadden (1994), Wooldridge (2010), and DasGupta (2008, p.243-244 for the general MLE problem) to assume that $\boldsymbol{x}_i$ is stochastic.

$$\sqrt{N}(\widehat{\boldsymbol{\beta}}_{mle} - \boldsymbol{\beta}_0) \rightarrow_d N(\boldsymbol{0}, \{\mathbb{E}[\frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1}) \iff \widehat{\boldsymbol{\beta}}_{mle} \sim_a N(\boldsymbol{\beta}_0, \{\mathbb{E}[\frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1}/N)$$

By replacing $\mathbb{E}(.)$ and $\boldsymbol{\beta}_0$, the estimated covariance matrix of $\widehat{\boldsymbol{\beta}}_{mle}$ is

$$\{N^{-1}\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}/N = \{\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}$$

in which $N^{-1}\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\boldsymbol{x}_i\boldsymbol{x}_i' \rightarrow_p \mathbb{E}[\frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']$. There are other estimators which also possess this property (Wooldridge, 2010, p.479-480).

Amemiya (1985, p.272-273), Agresti (2012, p.193), Cameron and Trivedi (2005, p.469), and Maddala (1983, p.25-26) assume that $\boldsymbol{x}_i$ is non-stochastic and state $\widehat{\boldsymbol{\beta}}_{mle} \sim_a N(\boldsymbol{\beta}_0, \{-lim_{N\rightarrow\infty}\frac{1}{N}\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]\}^{-1}/N) = N(\boldsymbol{\beta}_0, \{-\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]\}^{-1}) = N(\boldsymbol{\beta}_0, \{\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1})$. The estimated covariance matrix of $\widehat{\boldsymbol{\beta}}_{mle}$ is to replace $\boldsymbol{\beta}_0$ by $\widehat{\boldsymbol{\beta}}_{mle}$.

The covariance matrix can also be estimated by the bootstrapping method without relying on the formula above.

1.2.10 t-test

Under the null hypothesis $\beta_j = 0$,

$$t^* := \frac{\widehat{\beta}_{j,mle}}{\sqrt{[\{\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1}]_{j,j}}} \rightarrow_d N(0,1)$$

1.2.11 Quasi-Maximum Likelihood Estimation

According to White (1982) and Hansen (2022), a probability model is correctly specified if there exists at least one $\boldsymbol{\theta}' \in \Theta$ such that $g(z) = f(z;\boldsymbol{\theta}')$ for $\forall z$ where g is the true density (in the Data Generating Process DGP), f is the density specified by an econometrician i.e., g is in the parametric family of f. $\boldsymbol{\theta}'$ is then called the true population parameter and we denote it as $\boldsymbol{\theta}_0$. If for $\forall\boldsymbol{\theta} \in \Theta$, $g(z) \neq f(z;\boldsymbol{\theta})$ for at least one z, then the probability model is misspecified. By this definition, a correctly specified likelihood function does not necessarily use the true density if there exists $\boldsymbol{\theta}' \in \Theta$ such that the specified density reduces to the true density e.g., if g is Cauchy density, f is standard t density, and degree of freedom $\theta' = 1$.

White (1982) observes that $0 \leq KLIC(f,g;\boldsymbol{\theta}) = \mathbb{E}_g[ln\frac{g(z)}{f(z;\boldsymbol{\theta})}]$ for $\forall\boldsymbol{\theta} \in \Theta$ and defines the pseudo-true parameter as

$$\boldsymbol{\theta}^* := argmin_{\boldsymbol{\theta}\in\Theta}KLIC(f,g;\boldsymbol{\theta}) = argmax_{\boldsymbol{\theta}\in\Theta}\mathbb{E}_g[lnf(z;\boldsymbol{\theta})]$$

If the model is correctly specified i.e., $g(z) = f(z;\boldsymbol{\theta}_0)$ for at least one $\boldsymbol{\theta}_0 \in \Theta$, this implies $KLIC(f,g;\boldsymbol{\theta}_0) = 0 \implies \boldsymbol{\theta}^* = \boldsymbol{\theta}_0$. If the model is misspecified, $\boldsymbol{\theta}^*$ still exists, but $\boldsymbol{\theta}_0$ is not defined. White (1982) shows that under regularity conditions, the QMLE $\widehat{\boldsymbol{\theta}}_{qmle} \rightarrow_{a.s.} \boldsymbol{\theta}^*$, which is the minimizer of KLIC (the ignorance of the true density) or the maximizer of $\mathbb{E}_g[lnf(z;\boldsymbol{\theta})]$. The QMLE is asymptotically normally distributed with a sandwich covariance matrix.

In the case of binary outcome model, y_i must be Bernoulli distributed. The model is correctly specified if there exists at least one $\boldsymbol{\beta}_0 \in \Theta$ such that the true mass function $g(y_i|\boldsymbol{x}_i) = f(y_i|\boldsymbol{x}_i;\boldsymbol{\beta}_0) = F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i} = F(\boldsymbol{z}_i'\boldsymbol{\beta}_0)$ where F is specified by an econometrician (Note f here is not F'). The last equality holds if we define

$$\boldsymbol{z}_i = \begin{cases} \boldsymbol{x}_i & \text{if } y_i = 1 \\ -\boldsymbol{x}_i & \text{if } y_i = 0 \end{cases}$$

and assume F is symmetric at 0 i.e., $F(-a) = 1 - F(a)$. Such a transformation simplifies the proof of the properties of QMLE.

1.2.12 Consistency of Quasi-Maximum Likelihood Estimator

Theorem 6 (Theorem 1 of Abadie, Imbens and Zheng (2014) with $\psi(y_i, \boldsymbol{x}_i; \boldsymbol{\theta}) = \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$)
Suppose that $(y_i, \boldsymbol{x}_i')'$ is i.i.d. and (i) For some compact $\Theta \subset \mathbb{R}^K$, there is a unique value $\boldsymbol{\theta}^* \in \Theta$, such that $\mathbb{E}_g[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}^*}] = \mathbf{0}$. (ii) $\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ is continuous at each $\boldsymbol{\theta} \in \Theta$ with probability one. (iii) $\mathbb{E}_g[sup_{\boldsymbol{\theta}\in\Theta}||\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}||] < \infty$. Then, $\widehat{\boldsymbol{\theta}}_{qmle} \to_p \boldsymbol{\theta}^*$

Remark: Note that this theorem is a special case of the theorem in Newey and McFadden (1994) for the consistency of GMM estimator. $\mathbb{E}_g(.)$ is used to denote that the expectation is with respect to the true density. Also, the true population parameter $\boldsymbol{\theta}_0$ in the original theorem is changed to the pseudo-true parameter $\boldsymbol{\theta}^*$. Newey and McFadden (1994) suggest not analyzing the consistency of ML estimator by treating it as a special case of GMM estimator because it is rather difficult to provide primitive conditions for (i). The issue is usually solved by directly assuming (i). I solve this by assuming Θ to be convex, in addition to compactness, so that the theorems from the convex optimization can be applied.

Theorem 7 (Consistency of Logit QMLE)
Suppose that $(y_i, \boldsymbol{x}_i')'$ is i.i.d. and F is a logistic function and (i) Θ is compact and convex. (ii) $\mathbb{E}_g[\boldsymbol{x}_i\boldsymbol{x}_i']$ is a positive definite. (iii) $\mathbb{E}_g||\boldsymbol{x}_i|| < \infty$. (iv) Sufficient conditions for interchanging integration and differentiation. Then, $\widehat{\boldsymbol{\beta}}_{qmle} \to_p \boldsymbol{\beta}^*$.

Proof:
$lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ in Theorem 6 is $lnF(\boldsymbol{z}_i'\boldsymbol{\beta}) = ln\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})$ since Λ is symmetric at 0. (i) is for Theorem 6's (i).

$$\begin{aligned}
\boldsymbol{Q}(\boldsymbol{\beta}) &:= -\frac{\partial^2 \mathbb{E}_g[lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})]}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'} \\
&= -\frac{\partial^2 \mathbb{E}_g[lnF(\boldsymbol{z}_i'\boldsymbol{\beta})]}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'} \\
&= -\mathbb{E}_g[\frac{\partial^2 lnF(\boldsymbol{z}_i'\boldsymbol{\beta})]}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}] && \text{since (iv)} \\
&= -\mathbb{E}_g[\frac{\partial \frac{\partial lnF(\boldsymbol{z}_i'\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}}{\partial\boldsymbol{\beta}'}] \\
&= -\mathbb{E}_g[\frac{\partial \frac{\partial lnF(\boldsymbol{z}_i'\boldsymbol{\beta})}{\partial\boldsymbol{z}_i'\boldsymbol{\beta}}\frac{\partial\boldsymbol{z}_i'\boldsymbol{\beta}}{\partial\boldsymbol{\beta}}}{\partial\boldsymbol{\beta}'}] \\
&= -\mathbb{E}_g[\frac{\partial h(\boldsymbol{z}_i'\boldsymbol{\beta})\boldsymbol{z}_i}{\partial\boldsymbol{\beta}'}] \\
&= \mathbb{E}_g[-\frac{\partial h(\boldsymbol{z}_i'\boldsymbol{\beta})}{\partial\boldsymbol{z}_i'\boldsymbol{\beta}}\frac{\partial\boldsymbol{z}_i'\boldsymbol{\beta}}{\partial\boldsymbol{\beta}'}\boldsymbol{z}_i] \\
&= \mathbb{E}_g[H(\boldsymbol{z}_i'\boldsymbol{\beta})\boldsymbol{z}_i\boldsymbol{z}_i']
\end{aligned}$$

For the logit model i.e., $F = \Lambda$, $H(a) = \Lambda(a)(1-\Lambda(a)) = \Lambda(a)\Lambda(-a)$. Thus, $\boldsymbol{Q}(\boldsymbol{\beta}) = \mathbb{E}_g[\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})(1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}))\boldsymbol{z}_i\boldsymbol{z}_i'] = \mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta})(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}))\boldsymbol{x}_i\boldsymbol{x}_i']$ since Λ is symmetric at 0. $\boldsymbol{Q}(\boldsymbol{\beta}) = \mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta})(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}))\boldsymbol{x}_i\boldsymbol{x}_i']$ is a positive definite by (ii) as $\Lambda(a)(1-\Lambda(a)) > 0$ for any a. Thus $\mathbb{E}_g[lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})]$ is a strictly concave function of $\boldsymbol{\beta}$. On the convex domain, the first order condition $\mathbb{E}_g[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}^*}] = \frac{\partial\mathbb{E}_g[lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})]}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}^*} = \mathbf{0}$ becomes a sufficient condition (Theorem 7.15 of Sundaram (1996)) and there is a unique maximizer $\boldsymbol{\beta}^* \in \Theta$ (Theorem 7.14 of Sundaram (1996)) if it exists. This implies (i) of Theorem 6. $h = 1 - \Lambda$ for the logit model, thus

$$\begin{aligned}
\mathbb{E}_g[sup_{\boldsymbol{\beta}\in\Theta}||\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}||] &= \mathbb{E}_g[sup_{\boldsymbol{\beta}\in\Theta}||h(\boldsymbol{z}_i'\boldsymbol{\beta})\boldsymbol{z}_i||] = \mathbb{E}_g[sup_{\boldsymbol{\beta}\in\Theta}|1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})|||\boldsymbol{z}_i||] \\
&< \mathbb{E}_g||\boldsymbol{z}_i|| && \text{since } sup_{\boldsymbol{\beta}\in\Theta}|1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})| < 1 \\
&= \mathbb{E}_g||\boldsymbol{x}_i|| < \infty && \text{by (iii)}
\end{aligned}$$

This verifies (iii) of Theorem 6. $\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = \frac{\partial ln\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}$ is a continuous function of $\boldsymbol{\beta}$. This verifies (ii) of Theorem 6. Q.E.D.

Remark: Ruud (1983) shows that if $\mathbb{E}_g[x_j|z]$ is a linear function of $z = \alpha + \boldsymbol{x}'\boldsymbol{\beta}$ for $\forall j \in \{2,\cdots,p\}$ (e.g., if $\boldsymbol{x}$ follows multivariate normal), we have $\boldsymbol{\beta}^* = scalar \cdot \boldsymbol{\beta}_0$. Therefore, the QMLE converges in probability to a scaled true population parameter.

Interestingly, if the binary outcome model has only one independent variable $x \sim N(\mu_0, \sigma^2)$ if $y = 0$ and $x \sim N(\mu_1, \sigma^2)$ if $y = 1$, then $F(.)$ must be a logistic function (Agresti, 2012, p.169). Let $p = Pr(y = 1)$ and f_1 be the density of x given $y = 1$. By Bayes' Theorem,

$$Pr(y=1|x) = \frac{f_1 \cdot p}{f_1 \cdot p + f_0 \cdot (1-p)} = (\frac{f_1 \cdot p + f_0 \cdot (1-p)}{f_1 \cdot p})^{-1} = (1+\frac{(1-p)\cdot f_0}{p \cdot f_1})^{-1} = \frac{1}{1+exp(-[\underbrace{\frac{\mu_1-\mu_0}{\sigma^2}}_{\beta}x + \underbrace{\frac{\mu_0^2-\mu_1^2}{2\sigma^2} - ln\frac{1-p}{p}}_{\alpha}])}$$

$$\begin{aligned}
\frac{(1-p)}{p}\frac{f_0}{f_1} &= exp(ln\frac{1-p}{p})\frac{exp(-\frac{1}{2}(\frac{x-\mu_0}{\sigma})^2)}{exp(-\frac{1}{2}(\frac{x-\mu_1}{\sigma})^2)} \\
&= exp(ln\frac{1-p}{p})exp(-\frac{1}{2\sigma^2}(x^2 - 2x\mu_0 + \mu_0^2 - x^2 + 2x\mu_1 - \mu_1^2)) \\
&= exp(ln\frac{1-p}{p})exp(-\frac{1}{2\sigma^2}(-2x(\mu_0-\mu_1) + \mu_0^2 - \mu_1^2)) \\
&= exp(\frac{\mu_0-\mu_1}{\sigma^2}x - \frac{\mu_0^2-\mu_1^2}{2\sigma^2} + ln\frac{1-p}{p})
\end{aligned}$$

1.2.13 Asymptotic Normality of Quasi-Maximum Likelihood Estimator

Theorem 8 (Theorem 2 of Abadie, Imbens and Zheng (2014) with $\psi(y_i, \boldsymbol{x}_i; \boldsymbol{\theta}) = \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}$)
Suppose that $(y_i, \boldsymbol{x}_i')'$ is i.i.d. and the assumptions of Theorem 6 are satisfied and (i) $\boldsymbol{\theta}^*$ is an interior point of Θ. (ii) $\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}$ is continuously differentiable with respect to $\boldsymbol{\theta}$ in an open neighborhood $\mathbb{B}$ of $\boldsymbol{\theta}^*$. (iii) $\mathbb{E}_g[||\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}|_{\boldsymbol{\theta}^*}||^2] < \infty$. (iv) $\mathbb{E}_g[sup_{\boldsymbol{\theta}\in\mathbb{B}}||\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}||] < \infty$. (v) $\mathbb{E}_g[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}|_{\boldsymbol{\theta}^*}]$ is invertible. Then,

$$\sqrt{N}(\widehat{\boldsymbol{\theta}}_{qmle} - \boldsymbol{\theta}^*) \rightarrow_d N(\mathbf{0}, (\mathbb{E}_g[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}|_{\boldsymbol{\theta}^*}])^{-1}\mathbb{E}_g[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}|_{\boldsymbol{\theta}^*}(\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}|_{\boldsymbol{\theta}^*})']((\mathbb{E}_g[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}|_{\boldsymbol{\theta}^*}])^{-1})')$$

Theorem 9 (Asymptotic Normality of Logit QMLE)
Suppose that $(y_i, \boldsymbol{x}_i')'$ is i.i.d. and the assumptions of Theorem 7 are satisfied and (i) $\boldsymbol{\beta}^*$ is an interior point of Θ. (ii) $\mathbb{E}_g||\boldsymbol{x}_i||^4 < \infty$. Then,

$$\begin{aligned}
\sqrt{N}(\widehat{\boldsymbol{\beta}}_{qmle} - \boldsymbol{\beta}^*) &\rightarrow_d N(\mathbf{0}, (\mathbb{E}_g[H(\boldsymbol{z}_i'\boldsymbol{\beta}^*)\boldsymbol{z}_i\boldsymbol{z}_i'])^{-1}\mathbb{E}_g[h(\boldsymbol{z}_i'\boldsymbol{\beta}^*)^2\boldsymbol{z}_i\boldsymbol{z}_i']((\mathbb{E}_g[H(\boldsymbol{z}_i'\boldsymbol{\beta}^*)\boldsymbol{z}_i\boldsymbol{z}_i'])^{-1})') \\
&= N(\mathbf{0}, (\mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}^*)(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}^*))\boldsymbol{x}_i\boldsymbol{x}_i'])^{-1}\mathbb{E}_g[(y_i - \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}^*))^2\boldsymbol{x}_i\boldsymbol{x}_i']((\mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}^*)(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}^*))\boldsymbol{x}_i\boldsymbol{x}_i'])^{-1})')
\end{aligned}$$

Note that $(1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}))^2 = (y_i - \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}))^2$

Proof:

(i) is for Theorem 8's (i).

$$\begin{aligned}
\mathbb{E}_g[||\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}^*}||^2] &= \mathbb{E}_g[||h(\boldsymbol{z}_i'\boldsymbol{\beta}^*)\boldsymbol{z}_i||^2] \\
&= \mathbb{E}_g[|1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}^*)|^2||\boldsymbol{z}_i||^2] \\
&< \mathbb{E}_g||\boldsymbol{z}_i||^2 && \text{since } |1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}^*)|^2<1 \\
&= \mathbb{E}_g||\boldsymbol{x}_i||^2 < \infty && \text{implied by (ii)}
\end{aligned}$$

This verifies Theorem 8's (iii).

$$\begin{aligned}
\mathbb{E}_g[sup_{\boldsymbol{\beta}\in\mathbb{B}}||\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}||] &= \mathbb{E}_g[sup_{\boldsymbol{\beta}\in\mathbb{B}}||H(\boldsymbol{z}_i'\boldsymbol{\beta})\boldsymbol{z}_i\boldsymbol{z}_i'||] \\
&= \mathbb{E}_g[||\boldsymbol{z}_i\boldsymbol{z}_i'||sup_{\boldsymbol{\beta}\in\mathbb{B}}|H(\boldsymbol{z}_i'\boldsymbol{\beta})|] \\
&= \mathbb{E}_g[||\boldsymbol{x}_i\boldsymbol{x}_i'||sup_{\boldsymbol{\beta}\in\mathbb{B}}|\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})(1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}))|] \\
&< \mathbb{E}_g[||\boldsymbol{x}_i\boldsymbol{x}_i'||] && sup_{\boldsymbol{\beta}\in\mathbb{B}}|\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})(1-\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta}))|<1 \\
&\leq \mathbb{E}_g||\boldsymbol{x}_i||^4 < \infty && \text{by (ii)}
\end{aligned}$$

This verifies Theorem 8's (iv). Theorem 7's (ii) i.e., $\boldsymbol{Q}(\boldsymbol{\beta}) := -\mathbb{E}_g[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}]$ is positive definite for all $\boldsymbol{\beta}$. This means $\mathbb{E}_g[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}^*}]$ is a negative definite and thus invertible. This verifies Theorem 8's (v). $\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'} = \frac{\partial^2 ln\Lambda(\boldsymbol{z}_i'\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}$ exists and continuous, and thus Theorem 8's (ii) holds. Q.E.D.

Remark: The use of the sandwich estimator for the asymptotic covariance matrix means that you are applying QMLE, instead of MLE. If the model is correctly specified, $\boldsymbol{\beta}^* = \boldsymbol{\beta}_0$ and $\mathbb{E}_g[y_i|\boldsymbol{x}_i] = \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$. $\mathbb{E}_g[(y_i-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2\boldsymbol{x}_i\boldsymbol{x}_i'] = \mathbb{E}_g[\mathbb{E}_g[(y_i-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2|\boldsymbol{x}_i]\boldsymbol{x}_i\boldsymbol{x}_i'] = \mathbb{E}_g[Var_g(y_i|\boldsymbol{x}_i)\boldsymbol{x}_i\boldsymbol{x}_i'] = \mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i\boldsymbol{x}_i']$. Therefore,

$$\sqrt{N}(\widehat{\boldsymbol{\beta}}_{qmle}-\boldsymbol{\beta}_0) \to_d N(\boldsymbol{0}, (\mathbb{E}_g[\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i\boldsymbol{x}_i'])^{-1})$$

which is the usual MLE result.

1.3 Iteratively Weighted Non-linear Least Squares

1.3.1 Loss Function

$$\sum_{i=1}^{N} w_i(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2$$

1.3.2 First Order Condition

$$\begin{aligned}
\frac{\partial\sum_{i=1}^N w_i(y_i-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2}{\partial\boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{wnls}} &= \boldsymbol{0} \\
\sum_{i=1}^N w_i\frac{\partial(y_i-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2}{\partial\boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{wnls}} &= \boldsymbol{0} \\
-2\sum_{i=1}^N w_i(y_i-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{wnls}))\frac{\partial F(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{wnls}} &= \boldsymbol{0} \\
\sum_{i=1}^N w_iF'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{wnls})[y_i-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{wnls})]\boldsymbol{x}_i &= \boldsymbol{0}
\end{aligned}$$

If w_i is the inverse of the conditional variance of y_i i.e., $\frac{1}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})[1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})]}$, WNLS and ML estimation have the same FOC, except $\widehat{\boldsymbol{\beta}}$ in WNLS's weight is from NLS or the last step while that in MLE is still $\widehat{\boldsymbol{\beta}}_{mle}$.

1.3.3 Algorithm

For $t = 1 \cdots$

Calculate the weight w_i^t with $\widehat{\boldsymbol{\beta}}_{wnls}^{t-1}$ from last step or $\widehat{\boldsymbol{\beta}}_{nls}^{0}$ from NLS if $t = 1$.

Get $\widehat{\boldsymbol{\beta}}_{wnls}^{t} = arg\ min_{\boldsymbol{\beta}} \sum_{i=1}^{N} w_i^t (y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2 = arg\ min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (\sqrt{w_i^t} y_i - \sqrt{w_i^t} F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2$ with an appropriate algorithm for NLS e.g., Gauss-Newton method (see Appendix C for the details).

Repeat above two steps until the convergence of $\widehat{\boldsymbol{\beta}}_{wnls}^{t}$, which will be equal to $\widehat{\boldsymbol{\beta}}_{mle}$.

This algorithm is an application of the EM algorithm to the binary outcome model and thus it converges to the ML's solution. ML and NLS estimator are special cases of the M estimator, which is a special case of the extremum estimator (Amemiya, 1985). Therefore, under regularity conditions of the extremum estimator, NLS estimator is consistent and asymptotic normal. However, it is inefficient. WNLS with an inverse conditional variance weight is efficient. Thus, it is not necessary to apply IWNLS.

The following R code shows that IWNLS estimates of the logit model (F is a logistic function) converges to MLE.

```
binary_model_iwnls <- function (formula_, start_, data_, iter_ = 1000, tol_ = 1e-5) {
  b <- list()
  for (i in seq_len(iter_)) {
    if (i == 1) {
      m <-
        nls(
          as.formula(formula_),
          data = data_,
          start = start_
        )
    } else {
      m <-
        nls(
          as.formula(formula_),
          data = data_,
          start = start_,
          weights = w
        )
    }
    b[[i]] <- summary(m)$coefficients[ ,1]
    if (i > 1 && all(abs(b[[i]] - b[[i - 1]]) < tol_)) {
      return(list(model = m, number_of_iter = i))
    }
    p_hat <- predict(m, type = "response")
    w <- 1 / (p_hat * (1 - p_hat))
  }
  warning("not converge")
  return(list(model = m, number_of_iter = i))
}

logit_model_iwnls <-
  binary_model_iwnls(
    "y ~ 1 / (1 + exp(- b0 - b1 * x1 - b2 * x2))",
    start_ = list(b0 = 0.5, b1 = 0.5, b2 = 0.2),
    data_ = d
  )[["model"]]
```

```r
summary(logit_model_iwnls)

# check equivalence
# maximum likelihood estimation of logit model with newton method = iterative re-weighted least squares
summary(glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d))
```

1.4 GMM Estimation

1.4.1 Unconditional Moments

Set $w_i = \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}$,

$$\mathbb{E}[\frac{\partial w_i(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}_0}] = \mathbf{0}$$
$$\mathbb{E}[\underbrace{w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i}_{\boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta}_0)=\boldsymbol{g}_i}] = \mathbf{0}$$

It is just-identified because the number of moments i.e., $dim(\boldsymbol{x}_i) = dim(\boldsymbol{\beta}_0)$ i.e., the number of parameters. GMM reduced to MM estimation.

1.4.2 Asymptotic Distribution

$$\sqrt{N}(\widehat{\boldsymbol{\beta}}_{gmm} - \boldsymbol{\beta}_0) \to_d N(\mathbf{0}, (\boldsymbol{G}'\boldsymbol{W}\boldsymbol{G})^{-1}\boldsymbol{G}'\boldsymbol{W}\boldsymbol{S}\boldsymbol{W}\boldsymbol{G}(\boldsymbol{G}'\boldsymbol{W}\boldsymbol{G})^{-1})$$

where $\boldsymbol{G} = \mathbb{E}[\frac{\partial \boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]$ and $\boldsymbol{S} = \mathbb{E}[\boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta}_0)\boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta}_0)']$. $\boldsymbol{S}$ has this simple form because y_i is assumed to be independent.

$$\begin{aligned}
\boldsymbol{G} &= \mathbb{E}[\frac{\partial \boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] \\
&= \mathbb{E}[\frac{\partial w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta})(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}))\boldsymbol{x}_i}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] \\
&= \mathbb{E}[\mathbb{E}[\frac{\partial \frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta})}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}))\boldsymbol{x}_i}{\partial \boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}|\boldsymbol{x}_i]] \\
&= \mathbb{E}[-\frac{(F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']
\end{aligned}$$

$$\begin{aligned}
\boldsymbol{S} &= \mathbb{E}[\boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta}_0)\boldsymbol{g}(y_i,\boldsymbol{x}_i;\boldsymbol{\beta}_0)'] \\
&= \mathbb{E}[w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i(w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i)'] \\
&= \mathbb{E}[(w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2\mathbb{E}[(y_i - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2|\boldsymbol{x}_i]\boldsymbol{x}_i\boldsymbol{x}_i'] \\
&= \mathbb{E}[(w_i F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2 Var(y_i|\boldsymbol{x}_i)\boldsymbol{x}_i\boldsymbol{x}_i'] \\
&= \mathbb{E}[(\frac{F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))})^2 F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))\boldsymbol{x}_i\boldsymbol{x}_i'] \\
&= \mathbb{E}[\frac{(F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i'] = -\boldsymbol{G}
\end{aligned}$$

Because of the just-identification, GMM reduces to MM estimation, $\boldsymbol{G}$ is a square matrix and thus invertible.

$$\begin{aligned}
\sqrt{N}(\widehat{\boldsymbol{\beta}}_{mm} - \boldsymbol{\beta}_0) &\to_d N(\mathbf{0}, (\boldsymbol{G}'\boldsymbol{W}\boldsymbol{G})^{-1}\boldsymbol{G}'\boldsymbol{W}\boldsymbol{S}\boldsymbol{W}\boldsymbol{G}(\boldsymbol{G}'\boldsymbol{W}\boldsymbol{G})^{-1}) \\
&= N(\mathbf{0}, \boldsymbol{G}^{-1}\boldsymbol{W}^{-1}\boldsymbol{G}'^{-1}\boldsymbol{G}'\boldsymbol{W}\boldsymbol{S}\boldsymbol{W}\boldsymbol{G}\boldsymbol{G}^{-1}\boldsymbol{W}^{-1}\boldsymbol{G}'^{-1}) \\
&= N(\mathbf{0}, \boldsymbol{G}^{-1}\boldsymbol{S}\boldsymbol{G}'^{-1}) \\
&= N(\mathbf{0}, [\boldsymbol{I}(\boldsymbol{\beta}_0)]^{-1})
\end{aligned}$$

Because

$$\boldsymbol{G}^{-1}\boldsymbol{S}\boldsymbol{G}'^{-1} = -\boldsymbol{G}'^{-1} = \{\mathbb{E}[\frac{(F'(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2}{F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1} = [\boldsymbol{I}(\boldsymbol{\beta}_0)]^{-1}$$

Thus, GMM/MM and ML estimator has the same asymptotic distribution.

The following R code shows that GMM estimates of the logit model (F is a logistic function) are equal to that of MLE.

```
library(gmm)

logit_model_gmm <- function (y_, X_, init_par_) {

  if (!is.numeric(y_)) stop("y_ should be a numeric vector")
  if (!is.matrix(X_)) stop("X_ should be a matrix")
  if (!is.numeric(init_par_)) stop("init_par_ should be a numeric vector")
  if (length(init_par_) != ncol(X_) + 1) stop("wrong init_par_")

  x <- cbind(y_, X_)
  g <- function (theta, x) {
    y <- x[ ,1]
    X <- cbind(1, x[ ,-1])
    b <- theta[1:ncol(X)]
    p <- as.numeric(1 / (1 + exp(- X %*% b)))

    # the first derivative of logistic function is F(1 - F)
    # this leads to simplification i.e., cancel out the weight 1 / F(1 - F)
    (y - p) * X
  }

  # binary model is just-identified, thus G is a square matrix,
  # if G is invertible and W is chosen to be invertible e.g., I or S^-1,
  # the FOC of GMM reduces to the FOC of MM i.e.,
  # G(theta_hat)' W g(theta_hat) = 0 <=> g(theta_hat) = 0
  # therefore, default two-step with optimal weight matrix (W = S^-1 from first step)
  # and one-step with identity matrix (W = I) have the same FOC g(theta_hat) = 0
  # and thus yield the same estimates
  # note g is the sample average of g_i defined in note
  gmm(g = g, x = x, t0 = init_par_, wmatrix = "ident")
}

y <- d[["y"]]
X <- as.matrix(d[c("x1", "x2")])
summary(logit_model_gmm(y, X, c(0.5, 0.5, 0.2)))

# vs mle, same coefficient estimates
summary(glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d))
```

1.5 Bayesian Estimation

1.5.1 Joint Posterior Distribution

$$
\begin{aligned}
p(\boldsymbol{\theta}|\boldsymbol{y}, \boldsymbol{X}) &\propto L(\boldsymbol{y}|\boldsymbol{\theta}, \boldsymbol{X})\pi(\boldsymbol{\theta}|\boldsymbol{X}) \\
&= \prod_{i=1}^{N} F_{\boldsymbol{\theta}*}(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1 - F_{\boldsymbol{\theta}*}(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}\pi(\boldsymbol{\theta}|\boldsymbol{X})
\end{aligned}
$$

where $\boldsymbol{\theta} = (\boldsymbol{\beta}', \boldsymbol{\theta}^{*'})'$ and $\pi(\boldsymbol{\theta}|\boldsymbol{X})$ is the prior distribution of parameters.

1.5.2 Example: Bayesian Robit Model

$$
p(\boldsymbol{\beta}, \nu|\boldsymbol{y}, \boldsymbol{X}) \propto \prod_{i=1}^{N} F_{\nu}(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1 - F_{\nu}(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}\pi(\boldsymbol{\beta}, \nu|\boldsymbol{X})
$$

The closed form solution for the posterior mean $\mathbb{E}[p(\boldsymbol{\beta}, \nu|\boldsymbol{y}, \boldsymbol{X})]$ may not exist. Alternatively, the joint posterior distribution can be sampled with the Metropolis-Hastings algorithm (including Gibbs Sampler). The following R code demonstrates an example where $\pi(\boldsymbol{\beta}, \nu|\boldsymbol{X}) = \prod_j N(\beta_j|0, 10^2) \cdot Gamma(\nu|2, 0.1)$

```r
library(runjags)

model_data <-
  list(
    "y" = d[["y"]],
    "N" = length(d[["y"]]),
    "x1" = d[["x1"]],
    "x2" = d[["x2"]]
  )

robit_jags_model_string =
  "
    model {
      for (i in 1:N) {
        y[i] ~ dbern(p[i])
        p[i] <- pt(b[1] + b[2]*x1[i] + b[3]*x2[i], 0, 1, nu)
      }
      # prior
      for (j in 1:3) {
        b[j] ~ dnorm(0, 1 / 100)
      }
      nu ~ dgamma(2, 0.1)
    }
  "

robit_jags <-
  run.jags(
    model = robit_jags_model_string,
    n.chains = 4,
    data = model_data,
    monitor = c("b", "nu"),
    adapt = 1000,
    burnin = 5000,
    sample = 5000
  )
```

```
print(robit_jags)
plot(robit_jags, var = "b[1]")
plot(robit_jags, var = "b[2]")
plot(robit_jags, var = "b[3]")
plot(robit_jags, var = "nu")
```

https://discourse.mc-stan.org/t/robit-regression-not-robust/21245/5 includes R code for estimating Bayesian Robit model by using *brms* and *cmdstanr*.

Albert and Chib (1993) use the latent variable augmentation approach in which the conditional posterior distribution of the parameters is derived and estimate the Robit model with Gibbs Sampler. Koenker and Yoon (2009) provide a simulated comparison between MLE and Bayesian estimation of Robit / Gosset model with Albert and Chib (1993)'s method.

1.5.3 Bayesian GEV Model

Wang and Dey (2010) suggest using the distribution function of Generalized Extreme Value variable. Its response curve can be asymmetric.

```
library(runjags)

gev_jags_model_string =
  "
    model {
      for (i in 1:N) {
        y[i] ~ dbern(p[i])
        p[i] <- ifelse(xi == 0, exp(-exp(-(b[1] + b[2]*x1[i] + b[3]*x2[i]))), exp(-zz[i]^(-1 / xi)))
        zz[i] <- ifelse(z[i] < 0, 0, z[i])
        z[i] <- 1 + xi*(b[1] + b[2]*x1[i] + b[3]*x2[i])
      }
      # prior
      for (j in 1:3) {
        b[j] ~ dnorm(0, 1 / 100)
      }
      xi ~ dnorm(1, 1 / 100)
    }
  "

gev_jags <-
  run.jags(
    model = gev_jags_model_string,
    n.chains = 4,
    data = model_data,
    monitor = c("b", "xi"),
    adapt = 1000,
    burnin = 5000,
    sample = 5000
  )

print(gev_jags)
plot(gev_jags, var = "b[1]")
plot(gev_jags, var = "b[2]")
plot(gev_jags, var = "b[3]")
plot(gev_jags, var = "xi")
```

1.6 Special Case: Logit Model

If $F(.) = \Lambda(.)$ i.e., a logistic function (the c.d.f. of standard logistic random variable),

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}) = \frac{e^{\boldsymbol{x}_i'\boldsymbol{\beta}}}{1 + e^{\boldsymbol{x}_i'\boldsymbol{\beta}}} = \frac{1}{1 + e^{-\boldsymbol{x}_i'\boldsymbol{\beta}}}$$

$$ln\frac{p_i}{1 - p_i} = \Lambda^{-1}(p_i) = \boldsymbol{x}_i'\boldsymbol{\beta} \qquad \Lambda^{-1}(.) \text{ is called Logit function}$$

1.6.1 Marginal Effect

$$\begin{aligned}
\Lambda'(z) &= \frac{d(1 + e^{-z})^{-1}}{dz} \\
&= -(1 + e^{-z})^{-2} e^{-z}(-1) \\
&= (1 + e^{-z})^{-1}(1 + e^{-z})^{-1} e^{-z} \\
&= \Lambda(z)\frac{e^{-z}}{1 + e^{-z}} \\
&= \Lambda(z)\frac{1}{1 + e^{z}} \\
&= \Lambda(z)\frac{1 + e^{z} - e^{z}}{1 + e^{z}} \\
&= \Lambda(z)(1 - \Lambda(z))
\end{aligned}$$

Thus,

$$\Lambda'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{\beta} = \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta})(1 - \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}))\boldsymbol{\beta} \leq 0.25\boldsymbol{\beta}$$

Because

$$\begin{aligned}
\frac{dz(1 - z)}{dz}|_{z^*} &= 1 - 2z^* = 0 \\
z^* &= 1/2 \\
z^*(1 - z^*) &= 1/2 \cdot 1/2 = 1/4 = 0.25
\end{aligned}$$

$\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta})(1 - \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta})) = 0.25$ when $\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}) = 1/2$. This happens when $\boldsymbol{x}_i'\boldsymbol{\beta} = 0$ as p.d.f. of a standard logistic random variable is symmetric at 0 (c.d.f. = 0.5 at 0).

1.6.2 Odds Ratio

$$ln\frac{p_i}{1 - p_i} = \Lambda^{-1}(p_i) = \boldsymbol{x}_i'\boldsymbol{\beta}$$

If $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are different in x_k by 1 unit, then

$$\begin{aligned}
\boldsymbol{x}_i'\boldsymbol{\beta} - \boldsymbol{x}_j'\boldsymbol{\beta} &= ln\frac{p_i}{1 - p_i} - ln\frac{p_j}{1 - p_j} \\
\boldsymbol{x}_j'\boldsymbol{\beta} + 1 \cdot \beta_k - \boldsymbol{x}_j'\boldsymbol{\beta} = \beta_k &= ln\frac{p_i/(1 - p_i)}{p_j/(1 - p_j)} \\
exp(\beta_k) &= \frac{p_i/(1 - p_i)}{p_j/(1 - p_j)} := OR
\end{aligned}$$

Odds Ratio (OR) can be interpreted as

$$\frac{p_j}{1-p_j} = exp(\boldsymbol{x}_j'\boldsymbol{\beta})$$
$$exp(\boldsymbol{x}_i'\boldsymbol{\beta}) = exp(\boldsymbol{x}_j'\boldsymbol{\beta} + 1 \cdot \beta_k) = exp(\boldsymbol{x}_j'\boldsymbol{\beta})exp(\beta_k)$$
$$= \frac{p_j}{1-p_j} exp(\beta_k)$$

So, an unit increase in x_k means the odds $\frac{p_j}{1-p_j}$ is multiplied by the Odds Ratio (OR) $exp(\beta_k)$.

1.6.3 First Order Condition

$$\sum_{i=1}^{N} \frac{y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1-\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))} \Lambda'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i = \mathbf{0}$$
$$\sum_{i=1}^{N} \frac{y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1-\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))} \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1-\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i = \mathbf{0}$$
$$\sum_{i=1}^{N} (y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i = \mathbf{0}$$
$$\sum_{i=1}^{N} (y_i - \mathbb{E}(y_i|\boldsymbol{x}_i))\boldsymbol{x}_i = \mathbf{0}$$

This is similar to the first order condition of the OLS estimation of the linear model. Moreover, if the intercept is included in $\boldsymbol{x}_i$.

$$\sum_{i=1}^{N} (y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})) \cdot 1 = 0 \qquad \text{“residuals” sum to 0}$$
$$N^{-1} \sum_{i=1}^{N} \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) = \bar{y}$$

An interesting result, $\bar{y}$ is the percentage of one in the sample, which is equal to the average predicted probability of Logit Model.

1.7 Special Case: Skewed Logit Model

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = 1 - [1 + exp(\boldsymbol{x}_i'\boldsymbol{\beta})]^{-\alpha}$$

It reduces to the logit model if $\alpha = 1$. The extra parameter α is estimated simultaneously with $\boldsymbol{\beta}$.

```
skew_logit <- function (
    y_,
    X_,
    init_par_,
    intercept_ = TRUE,
    method_ = "BFGS"
) {

  if (!is.numeric(y_)) stop("y_ should be a numeric vector")
  if (!is.matrix(X_)) stop("X_ should be a matrix")
  if (intercept_) X_ <- cbind(1, X_)
  if (!is.numeric(init_par_)) stop("init_par_ should be a numeric vector")

  negative_lnL <- function (par, y, X) {
```

```
    beta <- par[1:ncol(X)]
    alpha <- par[ncol(X) + 1]
    cdf <- 1 - (1 + exp(X %*% beta))^(-alpha)
    -sum(y * log(cdf) + (1 - y) * log(1 - cdf))
  }

  optim(
    par = init_par_,
    fn = negative_lnL,
    y = y_, X = X_,
    method = method_,
    hessian = TRUE
  )
}
```

1.8 Special Case: Generalized Logit Model (Stukel, 1988)

$$Pr(y_i = 1|\boldsymbol{x}_i) = \frac{1}{1 + e^{-h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}} = \Lambda(h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta}))$$

where $h_{\boldsymbol{\alpha}}(.)$ is a strictly increasing non-linear function, which reduces to an identity function if $\boldsymbol{\alpha} = \boldsymbol{0}$. Thus, the model becomes the logit model if $\boldsymbol{\alpha} = \boldsymbol{0}$.

1.9 Special Case: Probit Model

If $F(.) = \Phi(.)$ i.e., the c.d.f. of a standard normal random variable,

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}) = \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}} \phi(z)dz$$

$$\Phi^{-1}(p_i) = \boldsymbol{x}_i'\boldsymbol{\beta} \qquad \Phi^{-1}(.) \text{ is called Probit function, no closed form}$$

1.9.1 Marginal Effect

$$\begin{aligned}
\Phi'(z) &= \frac{d\int_{-\infty}^{z} \phi(a)da}{dz} \\
&= \phi(z) && \text{by First Fundamental Theorem of Calculus} \\
&= \frac{1}{\sqrt{2\pi}} exp(-\frac{1}{2}z^2)
\end{aligned}$$

$$\begin{aligned}
\Phi'(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{\beta} &= \frac{1}{\sqrt{2\pi}} exp(-\frac{1}{2}(\boldsymbol{x}_i'\boldsymbol{\beta})^2)\boldsymbol{\beta} \\
&\leq \frac{1}{\sqrt{2\pi}} \cdot 1 \cdot \boldsymbol{\beta} \approx 0.4\boldsymbol{\beta} && \text{as } 0 < exp(z) \leq 1 \text{ if } z \leq 0
\end{aligned}$$

If there is only one independent variable, the marginal effect of the probit and logit model is both maximized at $x = -\alpha/\beta$. If both models have similar marginal effects at this point, we have $0.25\beta_{logit} \approx 0.4\beta_{probit}$. Thus, a rule of thumb is $\beta_{logit} \approx 1.6\beta_{probit}$.

1.9.2 First Order Condition

$$\sum_{i=1}^{N} \underbrace{\{\frac{\Phi'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{\Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - \Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\}}_{\widehat{w}_i}[y_i - \Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})]\boldsymbol{x}_i = \boldsymbol{0}$$

```r
summary(glm(y ~ x1 + x2, family = binomial(link = "probit"), data = d))
```

1.10 Special Case: Heteroskedastic Probit Model (Harvey, 1976)

If $F_i(.) = \Phi_{\sigma_i}(.)$ i.e., the c.d.f. of $N(0, \sigma_i^2)$ random variable,

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = \Phi_{\sigma_i}(\boldsymbol{x}_i'\boldsymbol{\beta}) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}/\sigma_i) = \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\sigma_i} \phi(z)dz$$

It is easy to verify the third equality with a change of the variable,

$$\begin{aligned}
\Phi_{\sigma_i}(\boldsymbol{x}_i'\boldsymbol{\beta}) &= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}} \frac{1}{\sigma_i}\phi(z/\sigma_i)dz \\
&= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\sigma_i} \frac{1}{\sigma_i}\phi(z^*)\frac{dz}{dz^*}dz^* && z^* = z/\sigma_i \\
&= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\sigma_i} \phi(z^*)dz^* = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}/\sigma_i)
\end{aligned}$$

Set $\sigma_i = exp(\boldsymbol{z}_i'\boldsymbol{\gamma})$ where $\boldsymbol{\gamma}$ does not have an intercept.

$$Pr(y_i = 1|\boldsymbol{x}_i, \boldsymbol{z}_i) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}/exp(\boldsymbol{z}_i'\boldsymbol{\gamma}))$$

It is the heteroskedasticity of the latent variable y_i^* or its error u_i, instead of the observed binary y_i (although it is also heteroskedastic, see Linear Probability Model (LPM) below).

```r
hetero_probit <- function (
    y_,
    X_,
    Z_,
    init_par_,
    intercept_ = TRUE,
    method_ = "BFGS"
) {

  if (!is.numeric(y_)) stop("y_ should be a numeric vector")
  if (!is.matrix(X_)) stop("X_ should be a matrix")
  if (!is.matrix(Z_)) stop("Z_ should be a matrix")
  if (intercept_) X_ <- cbind(1, X_)
  if (!is.numeric(init_par_)) stop("init_par_ should be a numeric vector")

  negative_lnL <- function (par, y, X, Z) {
    beta <- par[1:ncol(X)]
    gamma <- par[(ncol(X) + 1):(ncol(X) + ncol(Z))]
    cdf <- pnorm(X %*% beta, sd = exp(Z %*% gamma), lower.tail = TRUE, log.p = FALSE)
    -sum(y * log(cdf) + (1 - y) * log(1 - cdf))
  }

  optim(
    par = init_par_,
    fn = negative_lnL,
    y = y_, X = X_, Z = Z_,
    method = method_,
    hessian = TRUE
  )
}
```

1.11 Special Case: Robit / Gosset Model (Liu, 2004)

If $F(.) = F_{t,\nu,\tau}(.)$ i.e., the c.d.f. of a scaled t random variable (location parameter = 0, free scale parameter τ, free shape parameter ν),

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = F_{t,\nu,\tau}(\boldsymbol{x}_i'\boldsymbol{\beta}) = F_{t,\nu}(\boldsymbol{x}_i'\boldsymbol{\beta}/\tau) = \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\tau} t_\nu(z)dz$$

where for $z^2 < \nu$, $F_{t,\nu}(z) = 0.5 + z\frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}F_{2,1}(0.5, \frac{\nu+1}{2}, 1.5, -\frac{z^2}{\nu})$ and $F_{2,1}$ is the hypergeometric function. Moreover, for $z > 0$, $F_{t,\nu}(z) = 1 - 0.5I(\nu/2, 0.5, \nu/(z^2+\nu))$ where I is the regularized incomplete beta function. As $F_{t,\nu}(-z) = 1 - F_{t,\nu}(z)$, $F_{t,\nu}(z) = 0.5I(\nu/2, 0.5, \nu/(z^2+\nu))$ for $z \le 0$. For even ν, we always have closed form solutions (Shaw, 2006 p.47). It is easy to verify the third equality with a change of the variable,

$$\begin{aligned} F_{t,\nu,\tau}(\boldsymbol{x}_i'\boldsymbol{\beta}) &= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}} t_{\nu,\tau}(z)dz \\ &= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}} t_\nu(z/\tau)\frac{1}{\tau}dz && \text{by the definition of } t_\nu(.) \\ &= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\tau} t_\nu(z^*)\frac{1}{\tau}\frac{dz}{dz^*}dz^* && z^* = z/\tau \\ &= \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}/\tau} t_\nu(z^*)dz^* = F_{t,\nu}(\boldsymbol{x}_i'\boldsymbol{\beta}/\tau) \end{aligned}$$

$$F_{t,\nu}^{-1}(p_i) = \boldsymbol{x}_i'\boldsymbol{\beta}/\tau$$

Shaw (2006) provides several explicit formulas for $F_{t,\nu}^{-1}$ if ν is a small integer. Shaw (2006) also provides a power series formula for the general (may not be an integer) ν.

If $\tau = \sqrt{\frac{\nu-2}{\nu}}$, the variance of the scaled t random variable is 1. If this is the case, we estimate $\boldsymbol{\beta}$ and degree of freedom ν simultaneously and then find out τ by the equation. τ and ν can also be specified directly e.g., if $\tau = 1.5484$ and $\nu = 7$, $F_{t,\nu,\tau}(.)$ is close to the logistic function (Liu, 2004), and thus the robit model is close to the logit model in such a case. As a student's / standard t random variable (location parameter = 0, scale parameter $\tau = 1$) converges to a standard normal random variable when the degree of freedom $\nu \to \infty$, the robit model is also a generalization of the probit model.

The latent variable approach demonstrates that the robit model is robust to the outliers since u_i follows the standard t distribution and thus has fatter tails. For $1 < \nu \le 2$, $Var(u_i) = \infty$ and for $0 < \nu \le 1$, $Var(u_i)$ is not defined. Therefore, $\boldsymbol{\beta}$ is only identified up to scale for $0 < \nu \le 2$ (see the section for Latent Variable Models).

Numerical results show that the standard t cdf is not log-concave (Bagnoli and Bergstrom, 2005). The graph in Lemma 1 of this note also shows that the inverse Mills ratio of the standard t variable $\frac{dlnF_t(t)}{dt} = f_t(t)/F_t(t)$ with small degree of freedom can be increasing over some intervals of t. Thus, $\frac{d^2lnF_t(t)}{dt^2} \ge 0$ for some t. Therefore, $-\frac{\partial^2 lnL(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}$ may not be a positive definite and thus is not invertible. Newton's method may fail.

If ν is estimated with $\boldsymbol{\beta}$ simultaneously, the regularity conditions of MLE may not hold, and thus the resulting ML estimators may not be consistent and asymptotically normal. Following Koenker and Yoon (2009), it may be better to estimate $\boldsymbol{\beta}$ given a grid of ν. The result is a profiled likelihood function $L_p(\nu) = L(\widehat{\boldsymbol{\beta}}(\nu), \nu)$. $\widehat{\nu} = argmax_\nu L_p(\nu)$. In this case, ν is the parameter of interest while $\boldsymbol{\beta}$ is the nuisance parameters. Note that the profiled likelihood is not a true likelihood and thus can lead to biased estimates e.g., Neyman-Scott problem. Another approach is to use Bayesian estimation (see above sections). Maximizing the profile likelihood in this case is analogous to maximizing $-AIC(\nu) = 2lnL(\widehat{\boldsymbol{\beta}}(\nu), \nu) - 2dim(\boldsymbol{\beta})$ as the second term is not a function of ν.

The below R code shows the ML estimation of Robit model. Quasi-Newton's method, specifically BFGS, is the default method. One may instead specify a user-supplied t-link function with glm() in R (see Koenker, 2006), which uses Fisher's scoring / Iteratively Reweighted Least Squares (Appendix B) in the estimation.

```
set.seed(123)

n <- 5000
x1 <- rnorm(n)
x2 <- rnorm(n)
y <- as.numeric(0.5 + 0.5*x1 + 0.2*x2 + rt(n, df = 4) > 0)
d <- data.frame(y = y, x1 = x1, x2 = x2)

# robit in r
robit_mle <- function (
    y_,
    X_,
    init_par_,
    df_ = NULL,
    tau_ = 1,
    set_variance_one_ = FALSE,
    intercept_ = TRUE,
    method_ = "BFGS"
) {

  if (!is.numeric(y_)) stop("y_ should be a numeric vector")
  if (!is.matrix(X_)) stop("X_ should be a matrix")
  if (intercept_) X_ <- cbind(1, X_)
  if (!is.numeric(init_par_)) stop("init_par_ should be a numeric vector")
  if (is.null(df_) && (length(init_par_) != ncol(X_) + 1)) stop("wrong init_par_")
  if (!is.null(df_) && (length(init_par_) != ncol(X_))) stop("wrong init_par_")
  if (!is.numeric(tau_)) stop("tau_ should be a number")

  negative_lnL <- function (par, y, X) {
    beta <- par[1:ncol(X)]

    if (is.null(df_)) {
      df <- par[ncol(X) + 1]
    } else if (!is.na(df_) && is.numeric(df_) && df_ > 0) {
      df <- df_
    } else {
      stop("df_ should be positve real number")
    }

    if (!set_variance_one_) {
      t_cdf <- pt(X %*% beta / tau_, df = df, lower.tail = TRUE, log.p = FALSE)
    } else {
      t_cdf <- pt(X %*% beta / sqrt((df - 2) / df), df = df, lower.tail = TRUE, log.p = FALSE)
    }

    -sum(y * log(t_cdf) + (1 - y) * log(1 - t_cdf))
  }
```

```
  optim(
    par = init_par_,
    fn = negative_lnL,
    y = y_, X = X_,
    method = method_,
    hessian = TRUE
  )
}

y <- d[["y"]]
X <- as.matrix(d[c("x1", "x2")])

# set tau = 1, estimate beta and df (i.e., nu) simultaneously
robit_mle(y, X, c(0.5, 0.5, 0.2, 4))

# set tau = ((df - 2) / df)^(1 / 2), estimate beta and df simultaneously
robit_mle(y, X, c(0.5, 0.5, 0.2, 4), set_variance_one_ = TRUE)

# set tau = 1.5484 and df = 7, robit model approximates to logit model,
robit_mle(y, X, c(0.5, 0.5, 0.2), tau_ = 1.5484, df_ = 7)

# check equivalence
summary(glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d))

# set tau = 1 and df = very large number, robit model converges to probit model
robit_mle(y, X, c(0.5, 0.5, 0.2), df_ = 99999)

# check equivalence
summary(glm(y ~ x1 + x2, family = binomial(link = "probit"), data = d))
```

1.12 Special Case: Log-log Link Model or Gumbel Model

If $F(.)$ is the c.d.f. of the standard type 1 extreme value distribution (also called standard log-Weibull distribution or standard Gumbel distribution),

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = exp(-exp(-\boldsymbol{x}_i'\boldsymbol{\beta}))$$

$$-ln(-ln(p_i)) = \boldsymbol{x}_i'\boldsymbol{\beta}$$

Unlike the logit and the probit model, the response curve $Pr(y = 1|\boldsymbol{x})$ of the log-log link model is not symmetric at $p_i = 0.5$.

1.13 Special Case: Linear Probability Model (LPM)

If $F(.)$ is an identity function,

$$p_i := Pr(y_i = 1|\boldsymbol{x}_i) = \boldsymbol{x}_i'\boldsymbol{\beta}$$

However, it is unlikely that $F(.)$ is an identity function because the resulting predicted probability can be larger than 1 or smaller than 0. MLE's first order condition is

$$\sum_{i=1}^{N}\{\frac{1}{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}\}[y_i - \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}]\boldsymbol{x}_i = \boldsymbol{0}$$

There is a default heteroskedasticity problem. As shown before, $Var(y_i|\boldsymbol{x}_i) = p_i(1-p_i) = \boldsymbol{x}_i'\boldsymbol{\beta}(1-\boldsymbol{x}_i'\boldsymbol{\beta})$ which depends on i. Heteroskedasticity can be solved by using GLS estimation i.e., WLS estimation with an inverse conditional variance weight. First, use $\widehat{\boldsymbol{\beta}}_{ols}$ to get the weight $[\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})]^{-1}$ for $\forall i$. Second, get

$$\begin{aligned}
\widehat{\boldsymbol{\beta}}_{wls} &= argmin_{\boldsymbol{\beta}} \sum_{i=1}^{N} \{\frac{1}{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})}\}[y_i - \boldsymbol{x}_i'\boldsymbol{\beta}]^2 \\
&= argmin_{\boldsymbol{\beta}} \sum_{i=1}^{N} \{\frac{1}{\sqrt{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})}}\}^2[y_i - \boldsymbol{x}_i'\boldsymbol{\beta}]^2 \\
&= argmin_{\boldsymbol{\beta}} \sum_{i=1}^{N} [\frac{y_i}{\sqrt{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})}} - \frac{\boldsymbol{x}_i'}{\sqrt{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})}}\boldsymbol{\beta}]^2 \\
&= argmin_{\boldsymbol{\beta}} \sum_{i=1}^{N} [\tilde{y}_i - \tilde{\boldsymbol{x}}_i'\boldsymbol{\beta}]^2
\end{aligned}$$

Therefore, $\widehat{\boldsymbol{\beta}}_{wls} = (\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}_i')^{-1}\sum_{i=1}^{N}\tilde{\boldsymbol{x}}_i\tilde{y}_i$. Its first order condition is

$$\sum_{i=1}^{N} \{\frac{1}{\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols}(1-\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{ols})}\}[y_i - \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{wls}]\boldsymbol{x}_i = \boldsymbol{0}$$

If we loop the above process until the convergence with $\widehat{\boldsymbol{\beta}}_{wls}$ from the last step as the weight, the resulting IWLS estimate is equal to ML estimate. It is a special case of IWNLS = ML estimation derived in the previous section for the general model. If $\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}} \to 0$ or $\to 1$, the optimization may be unstable.

If $F(.)$ is truly an identity function, the OLS estimator is still unbiased and consistent. The default heteroskedasticity can also be tackled by using the robust standard error e.g., Eicker-Huber-White standard error (White, 1980). However, it is still inefficient. Angrist and Pischke (2009, p.93) provides some reasons why they prefer OLS with the robust standard error to WLS estimation e.g., the efficiency gains from WLS may not be realized if $F(.)$ is not truly an identity function and the variance estimates can be noisy.

```r
library(lmtest)
library(sandwich)

lpm <- lm(y ~ ., data = d)

# HC0 = White's estimator
coeftest(lpm, vcov = vcovHC(lpm, type = "HC0"))
```

Some scholars e.g., Angrist (2001, Section 2.1), Angrist and Pischke (2009) advocate using LPM e.g., if your target is to estimate Average Treatment Effect (ATE) and you have only one indicator regressor, the OLS slope estimate is equal to the estimated ATE under the independence assumption, which can be justified by the random assignment. This result holds no matter Y is binary, discrete or continuous. $\beta = \frac{Cov(Y,D)}{Var(D)} = \mathbb{E}(Y|D=1) - \mathbb{E}(Y|D=0)$. the second equality uses the law of iterated expectation in the derivation.

$$\begin{aligned}
\mathbb{E}(Y|D=1) - \mathbb{E}(Y|D=0) &= \mathbb{E}(Y_1|D=1) - \mathbb{E}(Y_0|D=0) \\
&= \mathbb{E}(Y_1) - \mathbb{E}(Y_0) && Y_1, Y_0 \perp D \text{ under random assignment} \\
&= \mathbb{E}(Y_1 - Y_0) = ATE
\end{aligned}$$

When there is only one indicator regressor, the CEF must be linear $\mathbb{E}(Y|D) = \mathbb{E}(Y|D=0) + [\mathbb{E}(Y|D=1) - \mathbb{E}(Y|D=0)]D$.

When there are covariates X with the indicator, Angrist and Pischke (2009, p.102-103) admit that CEF $\mathbb{E}(Y|X,D)$ is almost certainly non-linear and thus linear regression is very likely a wrong specification of CEF. However, linear regression still provides the best linear approximation of CEF. Because CEF has a causal interpretation, its MMSE linear approximation should also have a causal interpretation. Moreover, regression coefficients still have a causal interpretation (similar to ATT) if the covariate X is discrete and saturated. Imbens and Rubin (2015) formalize this in a theorem (7.1i) saying that the indicator coefficient of a linear model with covariates identifies ATE under random assignment and random sampling. Proof:

$$\begin{aligned}
Q(\alpha,\beta,\boldsymbol{\theta}) &:= \mathbb{E}[(Y_i-\alpha-\beta D_i-\boldsymbol{x}_i'\boldsymbol{\theta})^2]\\
&= \mathbb{E}[(Y_i-\alpha-\mathbb{E}[\boldsymbol{x}_i]'\boldsymbol{\theta}-\beta D_i-\boldsymbol{x}_i'\boldsymbol{\theta}+\mathbb{E}[\boldsymbol{x}_i]'\boldsymbol{\theta})^2]\\
&= \mathbb{E}[(Y_i-(\alpha+\mathbb{E}[\boldsymbol{x}_i]'\boldsymbol{\theta})-\beta D_i-(\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})^2]\\
&= \mathbb{E}[(Y_i-\widetilde{\alpha}-\beta D_i-(\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})^2]\\
&= \mathbb{E}[(Y_i-\widetilde{\alpha}-\beta D_i)^2]-\mathbb{E}[((\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})^2]-2\mathbb{E}[(Y_i-\widetilde{\alpha}-\beta D_i)((\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})]\\
&= \mathbb{E}[(Y_i-\widetilde{\alpha}-\beta D_i)^2]-\mathbb{E}[((\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})^2]-2\mathbb{E}[Y_i(\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta}] && \text{see below}
\end{aligned}$$

Thus, $\frac{\partial Q}{\partial \beta}|_{\beta^*}=\frac{\partial \mathbb{E}[(Y_i-\widetilde{\alpha}-\beta D_i)^2]}{\partial \beta}|_{\beta^*}=0$ reduces to the case without covariates. Therefore, as shown before, $\beta^*=ATE$.

$$\begin{aligned}
&\mathbb{E}[\widetilde{\alpha}((\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})]=\widetilde{\alpha}(\mathbb{E}[\boldsymbol{x}_i]-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta}=0\\
&\mathbb{E}[\beta D_i((\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])'\boldsymbol{\theta})]=\beta\mathbb{E}[D_i(\boldsymbol{x}_i-\mathbb{E}[\boldsymbol{x}_i])']\boldsymbol{\theta}=\beta(\mathbb{E}[D_i\boldsymbol{x}_i']-\mathbb{E}[D_i\mathbb{E}[\boldsymbol{x}_i]'])\boldsymbol{\theta}=\beta(\mathbb{E}[D_i\boldsymbol{x}_i']-\mathbb{E}[D_i]\mathbb{E}[\boldsymbol{x}_i]')\boldsymbol{\theta}=0
\end{aligned}$$

The last equality is due to the independence of D_i and $\boldsymbol{x}_i \implies \mathbb{E}[D_i\boldsymbol{x}_i']=\mathbb{E}[D_i]\mathbb{E}[\boldsymbol{x}_i]'$ given random assignment in the large sample. It may not hold in the small sample. The theorem holds no matter Y is binary or not. Moreover, it holds no matter the CEF $\mathbb{E}(Y|X,D)$ is linear or non-linear. In traditional Econometrics and Statistics, we always want to specify a correct model (but we can't and know that "all models are wrong, but some are useful"). However, if our target is just the causal effect, we can specify a wrong model regardless of the true CEF.

Average derivatives (Stoker, 1986) provide the average marginal effect without specifying $F(.)$. Define $\mathbb{E}[y|\boldsymbol{x}]=m(\boldsymbol{x})$,

$$\begin{aligned}
AME := \mathbb{E}[\frac{\partial m(\boldsymbol{x})}{\partial \boldsymbol{x}}] &= -\mathbb{E}[m(\boldsymbol{x})\frac{\partial ln(f(\boldsymbol{x}))}{\partial \boldsymbol{x}}] && \text{by generalized Information matrix equality}\\
&= -\mathbb{E}[\mathbb{E}[y|\boldsymbol{x}]\frac{f'(\boldsymbol{x})}{f(\boldsymbol{x})}]\\
&= -\mathbb{E}[\mathbb{E}[y\frac{f'(\boldsymbol{x})}{f(\boldsymbol{x})}|\boldsymbol{x}]]\\
&= -\mathbb{E}[y\frac{f'(\boldsymbol{x})}{f(\boldsymbol{x})}]
\end{aligned}$$

Thus,

$$\widehat{AME}=-N^{-1}\sum_{i=1}^{N}y_i\frac{\widehat{f}'(\boldsymbol{x}_i)}{\widehat{f}(\boldsymbol{x}_i)}$$

If $\boldsymbol{x}$ follow multivariate normal,

$$\begin{aligned}
AME &= -\mathbb{E}[y\cdot-(\boldsymbol{x}-\boldsymbol{\mu}_x)\boldsymbol{\Sigma}_x^{-1}]\\
&= \mathbb{E}[y(\boldsymbol{x}-\boldsymbol{\mu}_x)\{\mathbb{E}[(\boldsymbol{x}-\boldsymbol{\mu}_x)(\boldsymbol{x}-\boldsymbol{\mu}_x)']\}^{-1}]\\
&= \{\mathbb{E}[(\boldsymbol{x}-\boldsymbol{\mu}_x)(\boldsymbol{x}-\boldsymbol{\mu}_x)']\}^{-1}\mathbb{E}[(\boldsymbol{x}-\boldsymbol{\mu}_x)y]
\end{aligned}$$

Thus, it is an OLS estimator of regressing y on $\boldsymbol{x}-\boldsymbol{\mu}_x$. This result holds no matter y is binary or not. If your target is AME with multivariate normal $\boldsymbol{x}$, you can simply regress y on $\boldsymbol{x}-\boldsymbol{\mu}_x$, the estimated coefficients is estimated AME. Even $\boldsymbol{x}$ is not multivariate normal, AME can be non-parametrically estimated using $-N^{-1}\sum_{i=1}^{N} y_i \frac{\widehat{f}'(\boldsymbol{x}_i)}{\widehat{f}(\boldsymbol{x}_i)}$.

LPM can also be justified on its analogy with linear discriminant function (not linear discriminant analysis LDA). Set $\widehat{y}_i = 1(\widehat{Pr}(y_i = 1|\boldsymbol{x}_i) > threshold)$, where $\widehat{Pr}(y_i = 1|\boldsymbol{x}_i) = \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{lpm}$. If the threshold is set at $(\bar{\boldsymbol{x}}_1'\widehat{\boldsymbol{\beta}}_{lpm} + \bar{\boldsymbol{x}}_2'\widehat{\boldsymbol{\beta}}_{lpm})/2$, LPM provides the same prediction with linear discriminant function. Since linear discriminant function does not require any distributional or functional assumptions, even the true CEF is non-linear i.e., our specification is wrong, the prediction from LPM is still justified. Linear discriminant function is $\boldsymbol{x}_i'\widehat{\boldsymbol{\lambda}}$ where $\widehat{\boldsymbol{\lambda}} = arg\ max_{\boldsymbol{\lambda}} \frac{\boldsymbol{\lambda}'(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^2}{\boldsymbol{\lambda}'\boldsymbol{S}\boldsymbol{\lambda}}$. This gives $\widehat{\boldsymbol{\lambda}} = \boldsymbol{S}^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)$. Given new observation $\boldsymbol{x}_0$, it will be classify as group 1 if $\widehat{\boldsymbol{\lambda}}'\boldsymbol{x}_0 > \frac{1}{2}(\widehat{\boldsymbol{\lambda}}'\bar{\boldsymbol{x}}_1 + \widehat{\boldsymbol{\lambda}}'\bar{\boldsymbol{x}}_2)$. it can be shown $\widehat{\boldsymbol{\beta}}_{lpm} = \widehat{\boldsymbol{\lambda}}\frac{SSErr_{lpm}}{N_1+N_2-2}$ (see Maddala, 1983 for proof). Thus,

$$\begin{aligned}
\widehat{y}_i &= 1(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{lpm} > (\bar{\boldsymbol{x}}_1'\widehat{\boldsymbol{\beta}}_{lpm} + \bar{\boldsymbol{x}}_2'\widehat{\boldsymbol{\beta}}_{lpm})/2) \\
&= 1(\boldsymbol{x}_i'\widehat{\boldsymbol{\lambda}}\frac{SSErr_{lpm}}{N_1 + N_2 - 2)} > \frac{1}{2}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)'\widehat{\boldsymbol{\lambda}}\frac{SSErr_{lpm}}{N_1 + N_2 - 2}) \\
&= 1(\boldsymbol{x}_i'\widehat{\boldsymbol{\lambda}} > \frac{1}{2}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)'\widehat{\boldsymbol{\lambda}})
\end{aligned}$$

1.14 Model Evaluation

1.14.1 Accuracy

Accuracy $= N^{-1}\sum_{i=1}^{N} 1(1\{\widehat{Pr}(y_i = 1|\boldsymbol{x}_i) \geq threshold\} = y_i)$. The threshold is usually set at 0.5. If the data used to compute the accuracy are the data used to estimate the model, it is called in-sample accuracy. If new data are used, it is called out-of-sample accuracy.

```r
logit_model <- glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d)
p_hat <- predict(logit_model, type = "response")

mean(as.numeric(as.numeric(p_hat >= 0.5) == y), na.rm = TRUE)
```

Accuracy measure is easy to understand. However, it can be misleading when the dependent variable in the sample is unbalanced e.g., the overall accuracy is perfect but the sensitivity or specificity (defined below) can be very poor.

1.14.2 Confusion Matrix (Classification Table) and ROC curve

The table partitions the sample into True Positive (TP), True Negative (TN), False Positive (FP), False Negative (FN). Sensitivity $:= Pr(Predict\ is\ positive|Actual\ is\ positive) = \frac{Pr(Predict\ is\ positive \cap Actual\ is\ positive)}{Pr(Actual\ is\ positive)} = \frac{TP}{TP+FN}$. Specificity $:= Pr(Predict\ is\ negative|Actual\ is\ negative) = \frac{Pr(Predict\ is\ negative \cap Actual\ is\ negative)}{Pr(Actual\ is\ negative)} = \frac{TN}{TN+FP}$. We want a model having high sensitivity and specificity. However, sensitivity and specificity must be negatively related.

The ROC curve plots sensitivity $\frac{TP}{TP+FN}$ (y-axis) versus (1 - specificity) $\frac{FP}{TN+FP}$ (x-axis). The model that increases (1 - specificity) with a larger increase in sensitivity is superior i.e., the closer the curve is to the upper left corner, the better the model's fit. That is the reason why a larger area under ROC curve is prefered.

```r
library(pROC)

logit_model <- glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d)
p_hat <- predict(logit_model, type = "response")

table(
  actual = y,
  prediction = as.numeric(p_hat >= 0.5)
```

```
)

roc(y ~ x1 + x2, data = d, na.rm = TRUE, plot = TRUE)
```

1.14.3 Akaike Information Criterion (AIC)

Claeskens and Hjort (2008) provide a robust AIC formula for the generalized linear model.

$$\text{AIC} = 2 \cdot ln[L(\widehat{\boldsymbol{\beta}}_{mle})] - 2 \cdot \alpha$$

where $\alpha = \sum_{i=1}^{N} \mathbb{E}_g(y_i - \mathbb{E}_g[y_i|\boldsymbol{x}_i])\widehat{\theta}_i/\widehat{\eta}$. Since $\theta_i = \boldsymbol{x}_i'\boldsymbol{\beta}$ and $\eta = 1$ for Logit model, $\alpha = \sum_{i=1}^{N} \mathbb{E}_g(y_i - \overbrace{\mathbb{E}_g[y_i|\boldsymbol{x}_i]}^{Pr(y_i=1|\boldsymbol{x}_i)})\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}$. α can be estimated by the bootstrap method.

Estimate the model and get $Pr(\widehat{y_i = 1}|\boldsymbol{x}_i)$ for $\forall i$.
For $b = 1$ to B
  Draw y_i^* from Bernoulli distribution with parameters $p_i = Pr(\widehat{y_i = 1}|\boldsymbol{x}_i)$ for $\forall i$.
  Use bootstrapped dataset $\{y_i^*, \boldsymbol{x}_i'\}_{i=1}^{N}$ to estimate a logit model and get $\widehat{\boldsymbol{\beta}}_{mle}^*$.
  Calculate $A_b = \sum_{i=1}^{N}(y_i^* - Pr(\widehat{y_i = 1}|\boldsymbol{x}_i))\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}^*$.

Thus, we have B number of A_b for a model. AIC of this model is $2 \cdot ln[L(\widehat{\boldsymbol{\beta}}_{mle})] - 2 \cdot \sum_{b=1}^{B} A_b/B$. The original AIC formula sets α equaling to the number of parameter, which is easier to compute. However, it can be less accurate if the true model is not the specified model.

The following R code computes the above algorithm.

```
library(parallel)
library(foreach)
library(doParallel)

bootstrap_IC <- function (data__, B__ = 4000) {
  m <- glm(y ~ x1 + x2, data = data__, family = binomial(link = "logit"))
  p_hat <- predict(m, type = "response")

  get_A <- function (data___) {
    boot_y_star <- rbinom(nrow(m$model), 1, p_hat)
    boot_d <- cbind(y = boot_y_star, data___[c("x1", "x2")])
    boot_m <-
      glm(
        y ~ x1 + x2,
        data = boot_d,
        family = binomial(link = "logit")
      )

    t(boot_y_star - p_hat) %*%
      as.matrix(cbind(1, boot_d[c("x1", "x2")])) %*%
      as.matrix(boot_m$coefficients)
  }

  registerDoParallel(detectCores())
  A <-
```

```
    foreach (b = seq_len(B__)) %dopar% {
      get_A(data___ = data__)
    }
  stopImplicitCluster()

  alpha_hat <- mean(as.numeric(A), na.rm = TRUE)
  return(as.numeric(2 * logLik(m) - 2 * alpha_hat))
}

bootstrap_IC(d)
-1 * glm(y ~ x1 + x2, data = d, family = binomial(link = "logit"))$aic
```

1.14.4 McFadden's (1974) Pseudo-R Squares

$$\begin{aligned}
R^2_{mcf} &= 1 - \frac{lnL_{fit}}{lnL_0} \\
&= 1 - \frac{\sum_{i=1}^N \{y_i ln\widehat{p}_i + (1-y_i)ln(1-\widehat{p}_i)\}}{\sum_{i=1}^N \{y_i ln\bar{y} + (1-y_i)ln(1-\bar{y})\}} \\
&= 1 - \frac{\sum_{i=1}^N \{y_i ln\widehat{p}_i + (1-y_i)ln(1-\widehat{p}_i)\}}{(\sum_{i=1}^N y_i)ln\bar{y} + (N-\sum_{i=1}^N y_i)ln(1-\bar{y})\}} \\
&= 1 - \frac{\sum_{i=1}^N \{y_i ln\widehat{p}_i + (1-y_i)ln(1-\widehat{p}_i)\}}{N\bar{y}ln\bar{y} + N(1-\bar{y})ln(1-\bar{y})}
\end{aligned}$$

With a binary dependent variable, $lnL_0 \le lnL_{fit} \le 0$ means $0 \le R^2_{mcf} \le 1$. It is a negative percentage change of log likelihood values. It is the default Pseudo-R^2 in Stata command *logit* and *probit*.

```
logit_model <- glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d)
mc_fadden_r2 <- 1 - as.numeric(logLik(logit_model) / logLik(update(logit_model, formula = y ~ 1)))
```

1.14.5 Cox & Snell's (1970) Pseudo-R Squares

This is the default pseudo-R^2 in SAS proc *logistic*.

$$R^2_{cs} = 1 - exp(-\frac{2}{N}(lnL_{fit} - lnL_0)) = 1 - (\frac{L_0}{L_{fit}})^{2/N}$$

With a binary dependent variable, $lnL_0 \le lnL_{fit} \le 0$ means $0 \le R^2_{cs} \le 1 - exp(\overbrace{\frac{2}{N}lnL_0}^{\le 0}) < 1$. Thus, it is less than one (lnL_0 cannot be negative infinity). For the normal linear regression with homoscedasticity and known variance (not necessarily binary dependent variable), we have

$$\begin{aligned}
R^2_{cs} &= 1 - exp[-\frac{2}{N}(-\frac{1}{2\sigma^2})(\sum_{i=1}^N (y_i - \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2 - \sum_{i=1}^N (y_i - \bar{y})^2)] \\
&= 1 - exp[\frac{1}{N\sigma^2}(SSError - SSTotal)] = 1 - exp[\frac{1}{N\sigma^2}(-SSReg)] \\
&\approx \frac{1}{N\sigma^2}SSReg \approx \frac{1}{N \cdot \sum_{i=1}^N (y_i - \bar{y})^2/N} SSReg = \frac{SSReg}{SST} = R^2
\end{aligned}$$

It applies first order Taylor approximation centered at zero i.e., $1 - e^z \approx -z$. We also substitute σ^2 with $\widehat{\sigma}^2 = \sum_{i=1}^N (y_i - \bar{y})^2/N$

1.14.6 Cragg & Uhler's (1970) Pseudo-R Squares

Since Cox & Snell's R^2 is smaller than one, it is easy to adjust this by dividing its maximum value.

$$\begin{aligned} R^2_{cu} &= R^2_{cs}/(1 - exp(\frac{2}{N} lnL_0)) \\ &= \frac{1 - (\frac{L_0}{L_{fit}})^{2/N}}{1 - L_0^{2/N}} \end{aligned}$$

Many authors refer this as Nagelkerke's (1991) pseudo-R^2. However, it can at least date back to Cragg and Uhler (1970) published on The Canadian Journal of Economics. Maddala (1983) cited this correctly.

1.14.7 Pearson X Squares Test and Hosmer & Lemeshow (1980) Test

These tests are usually omitted in the Econometrics textbooks. Assuming there are J groups, the value of $\boldsymbol{x}$ are the same in each group,

$$X^2 = \sum_{j=1}^{J} \frac{(\sum_{i=1}^{n_j} y_{ij} - n_j \hat{p}_j)^2}{n_j \hat{p}_j (1 - \hat{p}_j)} \rightarrow_p \chi^2$$

However, if $\boldsymbol{x}$ are unique for all observations e.g., there is at least one continuous independent variable, X^2 does not converge to χ^2. The summand is the square of Pearson residual. The residual is the z-score of a binomial random variable, thus is asymptotic standard normal.

Hosmer & Lemeshow (1980) extend this idea for the dataset without same $\boldsymbol{x}$. Groups are formed by sorting the predicted probabilities $\hat{p}_i$,

$$HL = \sum_{j=1}^{J} \frac{(\sum_{i=1}^{n_j} y_{ij} - \sum_{i=1}^{n_j} \hat{p}_{ij})^2}{(\sum_{i=1}^{n_j} \hat{p}_{ij})(1 - \sum_{i=1}^{n_j} \frac{\hat{p}_{ij}}{n_j})}$$

The HL statistic does not converge in probability to χ^2 because the summand is not a square of asymptotic standard normal random variable. However, when all the $\boldsymbol{x}$ are unique, it is approximated by χ^2 with $J - 2$ degree of freedom. Different software may calculate different values of HL due to the treatment of tie for $\hat{p}_i$. The number of groups J is arbitrary, and different values of J may lead to a reverse result.

2 Latent Variable Models

2.1 Index Function Model

$$y_i^* = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i \qquad y_i^* \text{ is unobservable}$$

But we can observe y_i,

$$\begin{aligned}
y_i &= 1(y_i^* > 0) \\
\mathbb{E}(y_i|\boldsymbol{x}_i) &= 1 \cdot Pr(y_i = 1|\boldsymbol{x}_i) + 0 \cdot Pr(y_i = 0|\boldsymbol{x}_i) \\
&= Pr(y_i = 1|\boldsymbol{x}_i) \\
&= Pr(y_i^* > 0|\boldsymbol{x}_i) \\
&= Pr(\boldsymbol{x}_i'\boldsymbol{\beta} + u_i > 0|\boldsymbol{x}_i) \\
&= Pr(u_i > -\boldsymbol{x}_i'\boldsymbol{\beta}|\boldsymbol{x}_i) \\
&= Pr(u_i \leq \boldsymbol{x}_i'\boldsymbol{\beta}|\boldsymbol{x}_i) && \text{if } u_i \text{ is symmetric at } 0 \\
&= F_u(\boldsymbol{x}_i'\boldsymbol{\beta})
\end{aligned}$$

If u_i follows standard uniform, the model is LPM. If u_i follows standard logistic distribution, the model is the logit model. If u_i follows standard normal distribution, it is the probit model. If u_i follows scaled t distribution (location parameter = 0, free scale parameter τ), it is the robit model. For skewed models, we model $Pr(y_i = 1|\boldsymbol{x}_i) = 1 - F_u(-\boldsymbol{x}_i'\boldsymbol{\beta})$. If u_i follows standard type 1 extreme value distribution (standard log-Weibull distribution or standard gumbel distribution), it is the log-log link/gumbel model. If u_i follows type 2 generalized logistic distribution, it is the skewed logit model.

2.2 Identification of parameters

$$y_i = 1 \implies y_i^* > 0 \implies \boldsymbol{x}_i'\boldsymbol{\beta} + u_i > 0$$

However, for any constant $c > 0$,

$$\boldsymbol{x}_i'\boldsymbol{\beta} + u_i > 0 \iff \boldsymbol{x}_i'c\boldsymbol{\beta} + cu_i > 0$$

So, $\boldsymbol{\beta}$ is not identified with $y_i = 1(y_i^* > 0)$. Thus, we restrict $Var(u_i|\boldsymbol{x}_i)$ to identify $\boldsymbol{\beta}$. If u_i follows standard logistic distribution, $Var(u_i|\boldsymbol{x}_i) = \pi^2/3$. If u_i follows standard normal distribution, $Var(u_i|\boldsymbol{x}_i) = 1$. If u_i follows standard gumbel distribution, $Var(u_i|\boldsymbol{x}_i) = \pi^2/6$.

2.3 Additive Random Utility Model (ARUM)

$y = 0$ means option 0 is chosen. Utility obtained from this is U_0; $y = 1$ means option 1 is chosen. Utility obtained from this is U_1.

$$\begin{aligned}
U_0 &= V_0 + \varepsilon_0 && V_0 \text{ is a deterministic component of utility} \\
U_1 &= V_1 + \varepsilon_1
\end{aligned}$$

$$y = 1(U_1 > U_0)$$

$$
\begin{aligned}
\mathbb{E}(y|\boldsymbol{x}) &= 1 \cdot Pr(y=1|\boldsymbol{x}) + 0 \cdot Pr(y=0|\boldsymbol{x}) \\
&= Pr(y=1|\boldsymbol{x}) \\
&= Pr(U_1 > U_0|\boldsymbol{x}) \\
&= Pr(V_1 + \varepsilon_1 > V_0 + \varepsilon_0|\boldsymbol{x}) \\
&= Pr(V_1 - V_0 > \varepsilon_0 - \varepsilon_1|\boldsymbol{x}) \\
&= Pr(\varepsilon_0 - \varepsilon_1 < V_1 - V_0|\boldsymbol{x}) \\
&= F_{\varepsilon_0 - \varepsilon_1}(V_1 - V_0)
\end{aligned}
$$

2.3.1 Special Case: Logit Model

If ε_0 and ε_1 are independent and both follow standard type 1 extreme value distribution (standard log-weibull distribution or standard gumbel distribution). It can be shown that $\varepsilon_0 - \varepsilon_1$ follows standard logistic distribution i.e., $F_{\varepsilon_0-\varepsilon_1}(.) = \Lambda(.)$. It can also be shown by direct integration,

$$
\begin{aligned}
Pr(y=1|\boldsymbol{x}) &= Pr(\varepsilon_0 - \varepsilon_1 < V_1 - V_0|\boldsymbol{x}) \\
&= Pr(\varepsilon_0 < \varepsilon_1 + V_1 - V_0|\boldsymbol{x}) \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\varepsilon_1+V_1-V_0} f(\varepsilon_0,\varepsilon_1)\partial\varepsilon_0\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\varepsilon_1+V_1-V_0} f_{\varepsilon_0}(\varepsilon_0)f_{\varepsilon_1}(\varepsilon_1)\partial\varepsilon_0\partial\varepsilon_1 && \text{as independence} \\
&= \int_{-\infty}^{\infty} f_{\varepsilon_1}(\varepsilon_1)[\int_{-\infty}^{\varepsilon_1+V_1-V_0} f_{\varepsilon_0}(\varepsilon_0)\partial\varepsilon_0]\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f_{\varepsilon_1}(\varepsilon_1)[\int_{-\infty}^{\varepsilon_1+V_1-V_0} e^{-\varepsilon_0}exp(-e^{-\varepsilon_0})\partial\varepsilon_0]\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f_{\varepsilon_1}(\varepsilon_1)exp(-e^{-\varepsilon_0})|_{-\infty}^{\varepsilon_1+V_1-V_0}\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f_{\varepsilon_1}(\varepsilon_1)[exp(-e^{-(\varepsilon_1+V_1-V_0)}) - exp(-e^{--\infty})]\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f_{\varepsilon_1}(\varepsilon_1)exp(-e^{-(\varepsilon_1+V_1-V_0)})\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}exp(-e^{-\varepsilon_1})exp(-e^{-(\varepsilon_1+V_1-V_0)})\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}exp(-e^{-\varepsilon_1} - e^{-(\varepsilon_1+V_1-V_0)})\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}exp(-e^{-\varepsilon_1} - e^{-\varepsilon_1}e^{-(V_1-V_0)})\partial\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}exp(-e^{-\varepsilon_1}(1 + e^{-(V_1-V_0)}))\partial\varepsilon_1 \\
&= 1/(1+e^{-(V_1-V_0)}) && \text{as } \textstyle\int_{-\infty}^{\infty} ae^{-\varepsilon}exp(-ae^{-\varepsilon})d\varepsilon = 1 \\
&= \Lambda(V_1 - V_0)
\end{aligned}
$$

If $V_1 - V_0 = \boldsymbol{x}'\boldsymbol{\beta}$, $Pr(y=1|\boldsymbol{x}) = \Lambda(\boldsymbol{x}'\boldsymbol{\beta})$. It is the logit model.

2.3.2 Special Case: Probit Model

If ε_0 and ε_1 are multivariate (bivariate here) standard normally distributed, any linear combination of ε_0 and ε_1 also follow standard normal. So, $\varepsilon_0 - \varepsilon_1$ follows univariate standard normal. i.e., $F_{\varepsilon_0-\varepsilon_1}(.) = \Phi(.)$.

2.4 Endogeneous Probit Model

2.4.1 Only one continuous endogeneous variable

The model imposes strong distributional assumptions. It is basically a non-linear simultaneous equation model.

$$\begin{aligned}
y_i^* &= \boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma + u_i \\
z_i &= \boldsymbol{w}_i'\boldsymbol{\alpha} + \varepsilon_i && \text{First stage} \\
y_i &= 1(y_i^* > 0)
\end{aligned}$$

$$\begin{pmatrix} u_i \\ \varepsilon_i \end{pmatrix} \sim N(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho\sigma_\varepsilon \\ \rho\sigma_\varepsilon & \sigma_\varepsilon^2 \end{pmatrix})$$

ε_i is thus correlated with u_i when $\rho \neq 0$. Therefore, z_i is correlated with u_i and thus z_i is endogeneous. Thus, we can test the exogeneity of z_i by $H_0 : \rho = 0$. IV estimation cannot be applied. Maximum Likelihood Estimation can be applied though.

The log likelihood function is

$$\begin{aligned}
lnL &= \sum_{i=1}^{N} lnf(y_i, z_i|\boldsymbol{x}_i, \boldsymbol{w}_i) \\
&= \sum_{i=1}^{N} ln[f(y_i|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i)f(z_i|\boldsymbol{x}_i, \boldsymbol{w}_i)] \\
&= \sum_{i=1}^{N} \{ln[f(y_i|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i)] + ln[f(z_i|\boldsymbol{x}_i, \boldsymbol{w}_i)]\} \\
&= \sum_{i=1}^{N} \{ln[\Phi((2y_i - 1)\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}})] + ln[\phi(\frac{z_i - \boldsymbol{w}_i'\boldsymbol{\alpha}}{\sigma_\varepsilon})/\sigma_\varepsilon]\}
\end{aligned}$$

$$\begin{aligned}
lnf(y_i|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i) &= y_i lnPr(y_i = 1|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i) + (1 - y_i)ln(1 - Pr(y_i = 1|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i)) \\
&= y_i ln\Phi(\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}}) + (1 - y_i)ln(1 - \Phi(\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}})) \\
&= y_i ln\Phi(\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}}) + (1 - y_i)ln\Phi(-\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}}) \\
&= ln\Phi((2y_i - 1)\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}})
\end{aligned}$$

$$\begin{aligned}
Pr(y_i = 1|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i) &= Pr(y_i^* > 0|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i) \\
&= Pr(u_i \leq \boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma|z_i, \boldsymbol{x}_i, \boldsymbol{w}_i) \\
&= Pr(\nu_i \leq \boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma|\boldsymbol{x}_i, \boldsymbol{w}_i) && \text{where } \nu_i := u_i|\varepsilon_i \sim N((\rho\sigma_\varepsilon/\sigma_\varepsilon^2)\varepsilon_i, 1 - \rho^2\sigma_\varepsilon^2/\sigma_\varepsilon^2) \\
&= Pr(\frac{\nu_i - (\rho/\sigma_\epsilon)\varepsilon_i}{\sqrt{1-\rho^2}} \leq \frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)\varepsilon_i}{\sqrt{1-\rho^2}}|\boldsymbol{x}_i, \boldsymbol{w}_i) \\
&= Pr(Z \leq \frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}}|\boldsymbol{x}_i, \boldsymbol{w}_i) \\
&= \Phi(\frac{\boldsymbol{x}_i'\boldsymbol{\beta} + z_i\gamma - (\rho/\sigma_\epsilon)(z_i - \boldsymbol{w}_i'\boldsymbol{\alpha})}{\sqrt{1-\rho^2}})
\end{aligned}$$

2.4.2 Only one dummy endogeneous variable

Angrist (2001, Section 4) suggests several estimation strategies if the causal effect (potential outcome framework) is the target. The simplest one is to specify a linear model e.g. $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + D_i\gamma + u_i$ and use IV/2SLS estimation regardless of the binary nature of y_i. In his own words, "Although the constant-effects assumption is clearly unrealistic, in practice, more general estimation strategies often lead to similar average effects" (Angrist, 2001, p.8).

Another strategy is to find a MMSE approximation of Complier Causal Response Function (CCRF) by using weighted non-linear least squares to minimize $\mathbb{E}[\kappa_i(y_i - \Phi(\boldsymbol{x}_i'\boldsymbol{\beta} + D_i\gamma))^2]$ where κ_i is Abadie's Kappa. The treatement status difference of CCRF is conditional Local Average Treatment Effect (LATE).

2.5 Estimation of Change in Consumer Surplus

Small and Rosen (1981) shows that under certain assumptions, the change in expected welfare (consumer surplus) per person is

$$\frac{\Delta E}{N} = -\frac{1}{\lambda}\int_{W_1^0}^{W_1^f} \pi_1(W_1, W_2)dW_1$$

where λ is the marginal utility of income, $\pi_1(W_1, W_2) = Pr(\varepsilon_2 - \varepsilon_1 < W_1 - W_2)$, W_j is the deterministic utility of good j, ε_j is the stochastic utility of good j, W_j^0 is the deterministic utility of good j at quantity 0.

If ε_2 and ε_1 are independent and both follow standard Gumbel distribution, we have $\pi_1(W_1, W_2) = \frac{exp(W_1)}{exp(W_1)+exp(W_2)} = 1/(1+e^{-(W_1-W_2)}) = 1/(1+e^{-(\boldsymbol{x}_1-\boldsymbol{x}_2)'\boldsymbol{\beta}})$ i.e., Logit model. Thus,

$$\begin{aligned}
\frac{\Delta E}{N} &= -\frac{1}{\lambda}\int_{W_1^0}^{W_1^f} \frac{exp(W_1)}{exp(W_1)+exp(W_2)}dW_1 \\
&= -\frac{1}{\lambda} ln\sum_{j=1}^{2} exp(W_j)|_{W_1^0}^{W_1^f} \\
&= -\frac{1}{\lambda}\{ln\sum_{j=1}^{2} exp(W_j^f) - ln\sum_{j=1}^{2} exp(W_j^0)\}
\end{aligned}$$

Thus, the estimate of the change in expected consumer surplus is

$$-\frac{1}{\widehat{\lambda}}\{ln\sum_{j=1}^{2} exp(\boldsymbol{x}_j'\widehat{\boldsymbol{\beta}}^f) - ln\sum_{j=1}^{2} exp(\boldsymbol{x}_j'\widehat{\boldsymbol{\beta}}^0)\}$$

by estimating two Logit models and estimate λ separately.

2.6 Single Agent Dynamic Discrete Choice Model

This model is popular in Structural Econometrics and Structural Industrial Organization. There is a representative who is maximizing her utility overtime, certain assumptions are made on her decision making behaviors under uncertainty.

2.6.1 Stochastic Dynamic Programming

Under certain conditions (see Theorem 3.2 of Rust (1994)), v is the unique solution to the Bellman's equation of the stochastic dynamic programming problem,

$$\begin{aligned}
v(x,d) &= u(x,d) + \beta\int\int\{max_{d'\in D(y)}[v(y,d') + \varepsilon(d')]\}q(d\varepsilon|y)\pi(dy|x,d) \\
&= u(x,d) + \beta\int G[\{v(y,d'), d' \in D(y)\}|y]\pi(dy|x,d)
\end{aligned}$$

where the social surplus $G[\{u(x,d), d \in D(x)\}|x] := \int \{max_{d \in D(x)}[u(x,d) + \varepsilon(d)]\} q(d\varepsilon|x)$. It means that under certain conditions, v is the unique fixed point to the contraction mapping $\Psi : B \to B$,

$$\Psi(v)(x,d) = u(x,d) + \beta \int G[\{v(y,d'), d' \in D(y)\}|y] \pi(dy|x,d)$$

Under certain conditions (see Theorem 3.3 of Rust (1994)), the conditional choice probability is

$$P(d|x) = \frac{\partial G[\{v(x,d), d \in D(x)\}|x]}{\partial v(x,d)}$$

If $q(d\varepsilon|x)$ follows standard type 1 extreme value distribution i.e., standard gumbel distribution and set $D(x) = \{0,1\}$, then $G[\{v(x,d), d \in D(x)\}|x] = ln[\sum_{d \in D(x)} exp(v(x,d))]$ and thus we have the logit model,

$$P(d|x) = \frac{exp(v(x,d))}{\sum_{d' \in D(x)} exp(v(x,d'))}$$

where $v(x,d)$ is the unique fixed point to the contraction mapping $\Psi : B \to B$ defined above. we have

$$\Psi(v)(x,d) = u(x,d) + \beta \int ln[\sum_{d' \in D(y)} exp(v(y,d'))] \pi(dy|x,d)$$

2.6.2 Maximum Likelihood Estimation by Nested Fixed Point (NFP) Algorithm

The log likelihood function is

$$\begin{aligned} lnL(\boldsymbol{\theta}) &= ln[\prod_{t=1}^{T} \{\prod_{j=0}^{1} P(d_t = j|x_t; \boldsymbol{\theta})^{1(d_t=j)}\} \pi(x_t|x_{t-1}, d_{t-1}; \boldsymbol{\theta})] \\ &= \sum_{t=1}^{T} \{\sum_{j=0}^{1} 1(d_t = j) ln \frac{exp(v(x_t, j; \boldsymbol{\theta}))}{\sum_{k=0}^{1} exp(v(x_t, k; \boldsymbol{\theta}))}\} + \sum_{t=1}^{T} ln\pi(x_t|x_{t-1}, d_{t-1}; \boldsymbol{\theta}) \end{aligned}$$

The NFP algorithm (Rust, 1987) is computationally intensive nested loops.

Outer Loop:

Find $\boldsymbol{\theta}$ to maximize $lnL(\boldsymbol{\theta})$.

Inner Loop:

For each guess of $\boldsymbol{\theta}$, find v by solving the Bellman's equation $\Psi(v)(x,d) = u(x,d) + \beta \int ln[\sum_{d' \in D(y)} exp(v(y,d'))] \pi(dy|x,d)$.

3 Hypothesis Test

3.1 Lagrange Multiplier Test

$$\boldsymbol{S} = \begin{pmatrix} s_1(\widehat{\boldsymbol{\beta}})' \\ \vdots \\ s_N(\widehat{\boldsymbol{\beta}})' \end{pmatrix} \qquad \text{where } s_i(\widehat{\boldsymbol{\beta}}) = \frac{\partial ln l_i(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}}$$

$$\alpha_{LM} = N\frac{\mathbf{1}'\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1}}{\mathbf{1}'\mathbf{1}} := N \cdot R_u^2 \to_d \chi^2$$

$\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'$ is the hat matrix. Thus, $\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1}$ is an OLS predicted value of Outer-Product-of-the-Gradient (OPG) regression in which $\mathbf{1}$ is the dependent variable and $\boldsymbol{S}$ is regressors. The Sum of Squares Regression i.e., Explained Sum of Squares is

$$\begin{aligned}(\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1})'\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1} &= \mathbf{1}'\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1} \\ &= \mathbf{1}'\boldsymbol{S}(\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\mathbf{1}\end{aligned}$$

$\mathbf{1}'\mathbf{1}$ is Sum of Squares Total. Thus, α_{LM} is N times R_u^2. Note that the uncentered R-squared R_u^2 is equal to the usual R-squared R^2 if the model does not have an intercept.

3.1.1 Normality Test (Bera, Jarque and Lee, 1984) of Probit Model

If the density function of the error term u_i in the latent variable model is within the Pearson family, its c.d.f. is

$$Pr(u_i \leq t) = \Phi(t + \gamma_1 t^2 + \gamma_2 t^3)$$

where γ_1 controls the third moment i.e., skewness while γ_2 controls the fourth moment i.e., excess kurtosis. Clearly, u_i is standard normally distributed if both γ_1 and γ_2 are zero. This is the zero hypothesis.

Extended probit model becomes

$$Pr(y_i = 1|\boldsymbol{x}_i) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta} + \gamma_1(\boldsymbol{x}_i'\boldsymbol{\beta})^2 + \gamma_2(\boldsymbol{x}_i'\boldsymbol{\beta})^3)$$

Under zero hypothesis $\gamma_1 = \gamma_2 = 0$,

$$s_i(\begin{pmatrix} \widehat{\boldsymbol{\beta}}_{mle} \\ \widehat{\gamma}_{1,mle} \\ \widehat{\gamma}_{2,mle} \end{pmatrix}) = \begin{pmatrix} \frac{\partial ln l_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\gamma}_{1,mle},\widehat{\gamma}_{2,mle},\gamma_1=\gamma_2=0} \\ \frac{\partial ln l_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \gamma_1}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\gamma}_{1,mle},\widehat{\gamma}_{2,mle},\gamma_1=\gamma_2=0} \\ \frac{\partial ln l_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \gamma_2}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\gamma}_{1,mle},\widehat{\gamma}_{2,mle},\gamma_1=\gamma_2=0} \end{pmatrix} = \begin{pmatrix} \widehat{\varepsilon}_i^G \boldsymbol{x}_i \\ \widehat{\varepsilon}_i^G(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2 \\ \widehat{\varepsilon}_i^G(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^3 \end{pmatrix}$$

where $\widehat{\varepsilon}_i^G := \frac{y_i - \Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}{\Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-\Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})$ is the generalized residual. We know that matrix $\boldsymbol{S}$ has variables $\widehat{\varepsilon}_i^G\boldsymbol{x}_i'$, $\widehat{\varepsilon}_i^G(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2$ and $\widehat{\varepsilon}_i^G(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^3$. We can then regress 1 on these variables (no intercept). The usual R-squared R^2 of such model times N is α_{LM}, which is asymptotically Chi-squared distributed with two degree of freedom under zero hypothesis. If α_{LM} is large enough, we reject zero hypothesis i.e., reject normality of u_i and $F(.)$ should not be $\Phi(.)$.

Implementation in R.

```
probit_model <- glm(y ~ x1 + x2, family = binomial(link = "probit"), data = d)

xb_hat <- predict(probit_model, type = "link")
```

```
p_hat <- predict(probit_model, type = "response")
e_g <- (d[["y"]] - p_hat) * dnorm(xb_hat) / (p_hat * (1 - p_hat))

opg_d <-
  data.frame(
    y = 1,
    e_g_x = e_g * as.matrix(cbind(1, d[c("x1", "x2")])),
    e_g_xb_square = e_g * xb_hat^2,
    e_g_xb_cube = e_g * xb_hat^3
  )

alpha_lm <- nrow(opg_d) * summary(lm(y ~ . -1, data = opg_d))$r.squared
p_value <- pchisq(alpha_lm, df = 2, lower.tail = FALSE, ncp = 0, log.p = FALSE)
if (p_value < 0.01) print("reject gamma_1 = gamma_2 = 0 <=> reject u_i is standard normal")
```

3.1.2 Heteroskedasticity Test of Probit Model

Harvey (1976) proposes

$$Pr(y_i = 1|\boldsymbol{x}_i) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}/exp(\boldsymbol{z}_i'\boldsymbol{\gamma}))$$

where $\boldsymbol{z}_i$ does not have an intercept. Under zero hypothesis, $\boldsymbol{\gamma} = \boldsymbol{0}$,

$$s_i(\begin{pmatrix} \widehat{\boldsymbol{\beta}}_{mle} \\ \widehat{\boldsymbol{\gamma}}_{mle} \end{pmatrix}) = \begin{pmatrix} \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\gamma}}_{mle},\boldsymbol{\gamma}=\boldsymbol{0}} \\ \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\gamma}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\gamma}}_{mle},\boldsymbol{\gamma}=\boldsymbol{0}} \end{pmatrix} = \begin{pmatrix} \widehat{\varepsilon}_i^G \boldsymbol{x}_i \\ -\widehat{\varepsilon}_i^G \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}\boldsymbol{z}_i \end{pmatrix}$$

because

$$\frac{\partial \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}/exp(\boldsymbol{z}_i'\boldsymbol{\gamma}))}{\partial \boldsymbol{\gamma}}|_{\boldsymbol{\gamma}=0} = -\phi(\boldsymbol{x}_i'\boldsymbol{\beta}/exp(\boldsymbol{z}_i'\boldsymbol{\gamma}))\boldsymbol{x}_i'\boldsymbol{\beta} exp(\boldsymbol{z}_i'\boldsymbol{\gamma})^{-1}\boldsymbol{z}_i|_{\boldsymbol{\gamma}=0} = -\phi(\boldsymbol{x}_i'\boldsymbol{\beta})\boldsymbol{x}_i'\boldsymbol{\beta}\boldsymbol{z}_i$$

where $\widehat{\varepsilon}_i^G := \frac{y_i - \Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}{\Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-\Phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\phi(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})$ is the generalized residual. We know that matrix $\boldsymbol{S}$ has variables $\widehat{\varepsilon}_i^G \boldsymbol{x}_i'$, $-\widehat{\varepsilon}_i^G \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}\boldsymbol{z}_i'$. We can then regress 1 on these variables (no intercept). The usual R-squared R^2 of such model times N is α_{LM}, which is asymptotically Chi-squared distributed with $dim(\boldsymbol{z}_i)$ degree of freedom under zero hypothesis. If α_{LM} is large enough, we reject zero hypothesis i.e., reject homoskedasticity of u_i. Likelihood ratio test is not recommended here since the log-likelihood value of the unrestricted model is less easy to compute.

3.1.3 Omitted Variable Test

$$Pr(y_i = 1|\boldsymbol{x}_i) = F(\boldsymbol{x}_i'\boldsymbol{\beta} + \boldsymbol{z}_i'\boldsymbol{\gamma})$$

Under zero hypothesis, $\boldsymbol{\gamma} = \boldsymbol{0}$,

$$s_i(\begin{pmatrix} \widehat{\boldsymbol{\beta}}_{mle} \\ \widehat{\boldsymbol{\gamma}}_{mle} \end{pmatrix}) = \begin{pmatrix} \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\gamma}}_{mle},\boldsymbol{\gamma}=\boldsymbol{0}} \\ \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\gamma}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\gamma}}_{mle},\boldsymbol{\gamma}=\boldsymbol{0}} \end{pmatrix} = \begin{pmatrix} \widehat{\varepsilon}_i^G \boldsymbol{x}_i \\ \widehat{\varepsilon}_i^G \boldsymbol{z}_i \end{pmatrix}$$

where $\widehat{\varepsilon}_i^G := \frac{y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})$ is the generalized residual. We know that matrix $\boldsymbol{S}$ has variables $\widehat{\varepsilon}_i^G \boldsymbol{x}_i'$, $\widehat{\varepsilon}_i^G \boldsymbol{z}_i'$. We can then regress 1 on these variables (no intercept). The usual R-squared R^2 of such model times N is α_{LM}, which is asymptotically Chi-squared distributed with $dim(\boldsymbol{z}_i)$ degree of freedom under zero hypothesis. If α_{LM} is large enough, we reject zero hypothesis i.e., reject no omitted variables. Likelihood ratio test is not recommended here if $dim(\boldsymbol{z}_i)$ is large.

3.1.4 Test of Heaviness of Tails and Symmetry of Logit Model

$$Pr(y_i = 1|\boldsymbol{x}_i) = \frac{1}{1+e^{-h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}} = \Lambda(h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta}))$$

where $h_{\boldsymbol{\alpha}}(.)$ is a strictly increasing non-linear function. Under zero hypothesis, $\boldsymbol{\alpha} = \boldsymbol{0}$,

$$s_i(\begin{pmatrix} \widehat{\boldsymbol{\beta}}_{mle} \\ \widehat{\boldsymbol{\alpha}}_{mle} \end{pmatrix}) = \begin{pmatrix} \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\alpha}}_{mle},\boldsymbol{\alpha}=\boldsymbol{0}} \\ \frac{\partial lnl_i(\boldsymbol{\beta},\gamma_1,\gamma_2)}{\partial \boldsymbol{\alpha}}|_{\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\alpha}}_{mle},\boldsymbol{\alpha}=\boldsymbol{0}} \end{pmatrix} = \begin{pmatrix} \widehat{\varepsilon}_i^G \boldsymbol{x}_i \\ \widehat{\varepsilon}_i^G \frac{\partial h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0},\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\alpha}}_{mle}} \end{pmatrix}$$

because

$$\frac{\partial \Lambda(h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta}))}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0}} = \Lambda'(h_{\boldsymbol{\alpha}=\boldsymbol{0}}(\boldsymbol{x}_i'\boldsymbol{\beta}))\frac{\partial h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0}} = \Lambda'(\boldsymbol{x}_i'\boldsymbol{\beta})\frac{\partial h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0}}$$

where $\widehat{\varepsilon}_i^G := \frac{y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})}{\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\Lambda'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}) = y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})$ is the generalized residual. We know that matrix $\boldsymbol{S}$ has variables $\widehat{\varepsilon}_i^G\boldsymbol{x}_i'$, $\widehat{\varepsilon}_i^G(\frac{\partial h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0},\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\alpha}}_{mle}})'$. We can then regress 1 on these variables (no intercept). The usual R-squared R^2 of such model times N is α_{LM}, which is asymptotically Chi-squared distributed with $dim(\frac{\partial h_{\boldsymbol{\alpha}}(\boldsymbol{x}_i'\boldsymbol{\beta})}{\partial \boldsymbol{\alpha}}|_{\boldsymbol{\alpha}=\boldsymbol{0},\widehat{\boldsymbol{\beta}}_{mle},\widehat{\boldsymbol{\alpha}}_{mle}})$ degree of freedom under zero hypothesis. If α_{LM} is large enough, we reject zero hypothesis.

3.2 Wald Test

3.2.1 Nonlinearity of the coefficients

Under zero hypothesis $\boldsymbol{h}(\boldsymbol{\beta}) = \boldsymbol{0}$,

$$W := \boldsymbol{h}(\widehat{\boldsymbol{\beta}}_{mle})'[\frac{\partial \boldsymbol{h}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}_{mle}}\{\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{mle}))}\boldsymbol{x}_i\boldsymbol{x}_i']\}^{-1}\frac{\partial \boldsymbol{h}(\boldsymbol{\beta})'}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}_{mle}}]^{-1}\boldsymbol{h}(\widehat{\boldsymbol{\beta}}_{mle}) \to_d \chi^2$$

3.3 Likelihood Ratio (LR) Test

$$LR := 2(lnL_{ur} - lnL_r) \to_d \chi^2(\#\ of\ restrictions)$$

where ur stands for unrestricted. If H_0 is all coefficients (except the intercept) are zero,

$$LR := 2(lnL_{ur} - [N\bar{y}ln\bar{y} + N(1-\bar{y})ln(1-\bar{y})]) \to_d \chi^2_{dim(\boldsymbol{\beta})-1}$$

lnL_r is derived in the section of McFadden's R^2.

4 Generalized Linear Model (GLM)

4.1 Exponential Family of Distribution

4.1.1 Probability Mass Function

$$f(y|\theta,\eta) = a(y)exp[\frac{y\cdot\theta - b(\theta)}{\eta} + c(y,\eta)]$$

4.1.2 Moments

$\mathbb{E}[\frac{dlnf(y|\theta,\eta)}{d\theta}] = 0$ implies

$$\mathbb{E}(y) = b'(\theta)$$

$$Var(y) = b''(\theta)\eta$$

If $\theta = \boldsymbol{x}'\boldsymbol{\beta}$, we have $\mathbb{E}(y) = b'(\boldsymbol{x}'\boldsymbol{\beta})$. Thus, $b'(.)$ is the canonical mean function and $b'^{-1}(.)$ is the canonical link function.

4.1.3 Special Case: Binomial Distribution

$$\begin{aligned}
f(y|p) &= \begin{pmatrix} n \\ p \end{pmatrix} p^y(1-p)^{n-y} \\
&= exp[ln(\begin{pmatrix} n \\ p \end{pmatrix} p^y(1-p)^{n-y})] \\
&= exp[ln\begin{pmatrix} n \\ p \end{pmatrix} + ln(p^y(1-p)^{n-y})] \\
&= exp[ln\begin{pmatrix} n \\ p \end{pmatrix} + y\cdot ln(p) + (n-y)ln(1-p)] \\
&= exp[ln\begin{pmatrix} n \\ p \end{pmatrix} + y\cdot ln(p) + n\cdot ln(1-p) - y\cdot ln(1-p)] \\
&= exp[ln\begin{pmatrix} n \\ p \end{pmatrix} + y(ln(p) - ln(1-p)) + n\cdot ln(1-p)] \\
&= exp[y(ln\frac{p}{1-p}) - (-n\cdot ln(1-p)) + ln\begin{pmatrix} n \\ p \end{pmatrix}]
\end{aligned}$$

Thus, $\eta = 1$, $a(y) = 1$, $\theta = ln\frac{p}{1-p}$, $b(\theta) = -n \cdot ln(1-p)$, $c(y, \eta) = ln\begin{pmatrix} n \\ p \end{pmatrix}$

$$\theta = ln\frac{p}{1-p} = logit(p) \iff logistic(\theta) = p$$

$$\begin{aligned} b(\theta) &= -n \cdot ln(1-p) \\ &= -n \cdot ln(1 - logistic(\theta)) \\ &= -n \cdot ln(1 - \frac{1}{1+e^{-\theta}}) \\ &= -n \cdot ln(\frac{e^{-\theta}}{1+e^{-\theta}}) \\ &= -n \cdot ln(\frac{1}{1+e^{\theta}}) \\ &= n \cdot ln(1+e^{\theta}) \end{aligned}$$

$$b'(\theta) = \frac{d\ n \cdot ln(1+e^{\theta})}{d\theta} = n\frac{1}{1+e^{\theta}}e^{\theta} = n \cdot logistic(\theta)$$

Thus, if $\mathbb{E}[\frac{dlnf(y|\theta,\eta)}{d\theta}] = 0$, we have $\mathbb{E}(y) = b'(\theta) = n \cdot logistic(\theta)$. Bernoulli distribution is a special case of Binomial distribution with $n = 1$. Therefore, binary outcome model, i.e., y is Bernoulli distributed, is $\mathbb{E}(y) = logistic(\boldsymbol{x}'\boldsymbol{\beta})$ by setting $n = 1$ and $\theta = \boldsymbol{x}'\boldsymbol{\beta}$. Logistic function is the canonical mean function for the binary outcome model. $F(.)$ is thus selected to be logistic function.

5 Conditional Logistic Regression

$$Pr(y_i = 1) = logistic(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta}) = \frac{e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}}{1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}}$$

5.1 Full Likelihood Function

$$\begin{aligned}
L(\boldsymbol{y}; \boldsymbol{\beta}, \gamma) &= \prod_{i=1}^{N} \frac{e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}} && \text{assume independence of } y_i \\
&= \frac{\prod_{i=1}^{N} e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\prod_{i=1}^{N}[1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}]} \\
&= \frac{e^{\sum_{i=1}^{N}(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\prod_{i=1}^{N}[1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}]} \\
&= \frac{e^{\gamma \sum_{i=1}^{N} y_i + \sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\prod_{i=1}^{N}[1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}]}
\end{aligned}$$

5.2 Conditional Likelihood Function

Define the set $G_t = \{\boldsymbol{y} \in \mathbb{R}^{dim(\boldsymbol{y})}; \boldsymbol{y}'\boldsymbol{1} = t\}$

$$\begin{aligned}
L(\boldsymbol{y}|\boldsymbol{y}'\boldsymbol{1} = t; \boldsymbol{\beta}, \gamma) &= \frac{Pr(\boldsymbol{y} \cap \boldsymbol{y}'\boldsymbol{1} = t)}{Pr(\boldsymbol{y}'\boldsymbol{1} = t)} && \text{conditional probability} \\
&= \frac{Pr(\boldsymbol{y} \cap \boldsymbol{y}'\boldsymbol{1} = t)}{Pr(\cup_{\boldsymbol{z} \in \mathbb{R}^{dim(\boldsymbol{z})}}\{\boldsymbol{z} \cap \boldsymbol{z}'\boldsymbol{1} = t\})} && \text{total probability} \\
&= \frac{Pr(\boldsymbol{y} \cap \boldsymbol{y}'\boldsymbol{1} = t)}{\sum_{\boldsymbol{z} \in \mathbb{R}^{dim(\boldsymbol{z})}} Pr(\boldsymbol{z} \cap \boldsymbol{z}'\boldsymbol{1} = t)} && \text{no intersection of sets} \\
&= \frac{Pr(\boldsymbol{y} \cap \boldsymbol{y}'\boldsymbol{1} = t)}{\sum_{\boldsymbol{z} \in G_t} Pr(\boldsymbol{z} \cap \boldsymbol{z}'\boldsymbol{1} = t)} \\
&= \frac{\frac{e^{\gamma t + \sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\prod_{i=1}^{N}[1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}]}}{\sum_{\boldsymbol{z} \in G_t} \frac{e^{\gamma t + \sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})z_i}}{\prod_{i=1}^{N}[1 + e^{(\gamma + \boldsymbol{x}_i'\boldsymbol{\beta})}]}} \\
&= \frac{e^{\gamma t + \sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\sum_{\boldsymbol{z} \in G_t} e^{\gamma t + \sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})z_i}} \\
&= \frac{e^{\gamma t} e^{\sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{e^{\gamma t} \sum_{\boldsymbol{z} \in G_t} e^{\sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})z_i}} \\
&= \frac{e^{\sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})y_i}}{\sum_{\boldsymbol{z} \in G_t} e^{\sum_{i=1}^{N}(\boldsymbol{x}_i'\boldsymbol{\beta})z_i}}
\end{aligned}$$

Parameter γ is gone.

6 Existence of MLE in Logit Model

The below theorem is a simplification of Theorem 1 in Albert and Anderson (1984). If the complete separation occurs, at least one estimated coefficients will become infinity and the estimated standard errors will become unnaturally high. Categorical variables can lead to the problem of complete separation if the responses of one of the categories are all one or all zero.

Theorem

If there is complete separation i.e., if there exists $\boldsymbol{\beta}$ such that $\boldsymbol{x}_i'\boldsymbol{\beta} > 0$ for i such that $y_i = 1$; $\boldsymbol{x}_i'\boldsymbol{\beta} < 0$ for i such that $y_i = 0$. Then, the Maximum Likelihood estimates of $\boldsymbol{\beta}$ in Logit model do not exist i.e., not finite.

Proof: For $k > 0$, let $\boldsymbol{\beta}^* = k\boldsymbol{\beta}$, $lnL(k\boldsymbol{\beta}) = \sum_{i=1}^{N} ln\frac{exp(y_i\boldsymbol{x}_i'k\boldsymbol{\beta})}{1+exp(\boldsymbol{x}_i'k\boldsymbol{\beta})}$. For i such that $y_i = 1$, $\boldsymbol{x}_i'\boldsymbol{\beta} > 0 \implies -\boldsymbol{x}_i'k\boldsymbol{\beta} \to -\infty$ as $k \to \infty \implies \sum_{i:y_i=1} ln\frac{1}{1+exp(-\boldsymbol{x}_i'k\boldsymbol{\beta})} = 0$ as $k \to \infty$. For i such that $y_i = 0$, $\boldsymbol{x}_i'\boldsymbol{\beta} < 0 \implies \boldsymbol{x}_i'k\boldsymbol{\beta} \to -\infty$ as $k \to \infty \implies \sum_{i:y_i=0} ln\frac{1}{1+exp(\boldsymbol{x}_i'k\boldsymbol{\beta})} = 0$ as $k \to \infty$. Thus, lnL reaches its highest possible value 0 when $k \to \infty$. Maximizing $lnL(.) \implies k \to \infty \implies \boldsymbol{\beta}^* \to \infty$ or $\to -\infty$. Q.E.D.

Remark: the complete separation condition means there is a separating hyperplane perfectly separating the observations. This can be checked visually with at most 3 explanatory variables for human being. If $Pr(\widehat{y_i = 1}|\boldsymbol{x}_i) = 1$ for $\forall y_i = 1$ and $Pr(\widehat{y_i = 1}|\boldsymbol{x}_i) = 0$ for $\forall y_i = 0$ or the log-likelihood is zero or the estimated covariance matrix is extremely large, there may be complete separation (Agresti, 2012).

7 Berkson's Minimum Chi-square Estimator

$$p_t := Pr(y_{it} = 1|\boldsymbol{x}_{it} = \boldsymbol{x}_t) = F(\boldsymbol{x}_t'\boldsymbol{\beta})$$
$$F^{-1}(p_t) = \boldsymbol{x}_t'\boldsymbol{\beta}$$

p_t can be estimated by $\bar{p}_t = \bar{y}_t = N_t^{-1}\sum_{i=1}^{N_t} y_{it}$

$$F^{-1}(\bar{p}_t) - F^{-1}(\bar{p}_t) + F^{-1}(p_t) = \boldsymbol{x}_t'\boldsymbol{\beta}$$
$$F^{-1}(\bar{p}_t) = \boldsymbol{x}_t'\boldsymbol{\beta} + \underbrace{F^{-1}(\bar{p}_t) - F^{-1}(p_t)}_{v_t}$$

First order Taylor expansion of $F^{-1}(\bar{p}_t)$ centered at p_t,

$$F^{-1}(\bar{p}_t) \approx F^{-1}(p_t) + \frac{dF^{-1}(\bar{p}_t)}{d\bar{p}_t}|_{p_t}(\bar{p}_t - p_t) = F^{-1}(p_t) + \frac{1}{f(p_t)}(\bar{p}_t - p_t) \iff v_t = F^{-1}(\bar{p}_t) - F^{-1}(p_t) = \frac{\bar{p}_t - p_t}{f(p_t)}$$
$$Var(v_t) \approx Var(\frac{\bar{p}_t - p_t}{f(p_t)}) = \frac{1}{(f(p_t))^2}Var(\bar{p}_t) = \frac{1}{(f(p_t))^2}\frac{p_t(1-p_t)}{N_t} = \frac{p_t(1-p_t)}{(f(p_t))^2N_t}$$

Since $Var(\bar{p}_t) = N_t^{-2}Var(\sum_{i=1}^{N_t} y_{it}) = N_t^{-2} \cdot N_t p_t(1-p_t) = \frac{p_t(1-p_t)}{N_t}$.

As $Var(v_t|\boldsymbol{x}_t)$ depends on t, there is heteroskedasticity. GLS (WLS here) can be used to estimate $\boldsymbol{\beta}$ efficiently.

$$\widehat{\boldsymbol{\beta}}_{chi} = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} Var(v_t|\boldsymbol{x}_t)^{-1}(F^{-1}(\bar{p}_t) - \boldsymbol{x}_t'\boldsymbol{\beta})^2$$

7.1 Special Case: F is an identity function

$$\mathbb{E}(v_t) = \mathbb{E}(\bar{p}_t - p_t) = N_t^{-1}\mathbb{E}(\sum_{i=1}^{N_t} y_{it}) - p_t = N_t^{-1} \cdot N_t p_t - p_t = 0$$
$$Var(v_t) = \frac{p_t(1-p_t)}{1^2N_t} = \frac{p_t(1-p_t)}{N_t}$$
$$\widehat{\boldsymbol{\beta}}_{chi} = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} \frac{N_t}{\bar{p}_t(1-\bar{p}_t)}(\bar{p}_t - \boldsymbol{x}_t'\boldsymbol{\beta})^2 = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} [\sqrt{\frac{N_t}{\bar{p}_t(1-\bar{p}_t)}}\bar{p}_t - (\sqrt{\frac{N_t}{\bar{p}_t(1-\bar{p}_t)}}\boldsymbol{x}_t)'\boldsymbol{\beta}]^2$$

It is called minimum chi-square method.

7.2 Special Case: F is a logistic function

$$Var(v_t) \approx \frac{p_t(1-p_t)}{[p_t(1-p_t)]^2N_t} = \frac{1}{N_t p_t(1-p_t)}$$
$$\widehat{\boldsymbol{\beta}}_{chi} = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} N_t\bar{p}_t(1-\bar{p}_t)(logit(\bar{p}_t) - \boldsymbol{x}_t'\boldsymbol{\beta})^2 = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} [\sqrt{N_t\bar{p}_t(1-\bar{p}_t)}logti(\bar{p}_t) - (\sqrt{N_t\bar{p}_t(1-\bar{p}_t)}\boldsymbol{x}_t)'\boldsymbol{\beta}]^2$$

It is called minimum logit chi-square method. It can be shown

$$\sqrt{T}(\widehat{\boldsymbol{\beta}}_{chi} - \boldsymbol{\beta}_0) \to_d N(\boldsymbol{0}, [\sum_{t=1}^{T} \frac{N_t exp(\boldsymbol{x}_t'\boldsymbol{\beta}_0)}{(1+exp(\boldsymbol{x}_t'\boldsymbol{\beta}_0))^2}\boldsymbol{x}_t\boldsymbol{x}_t']^{-1})$$

7.3 Special Case: F is standard normal cdf

$$Var(v_t) \approx \frac{p_t(1-p_t)}{(\phi(p_t))^2N_t}$$
$$\widehat{\boldsymbol{\beta}}_{chi} = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} \frac{(\phi(\bar{p}_t))^2N_t}{\bar{p}_t(1-\bar{p}_t)}(\Phi^{-1}(\bar{p}_t) - \boldsymbol{x}_t'\boldsymbol{\beta})^2 = arg\ min_{\boldsymbol{\beta}} \sum_{t=1}^{T} [\sqrt{\frac{(\phi(\bar{p}_t))^2N_t}{\bar{p}_t(1-\bar{p}_t)}}\Phi^{-1}(\bar{p}_t) - (\sqrt{\frac{(\phi(\bar{p}_t))^2N_t}{\bar{p}_t(1-\bar{p}_t)}}\boldsymbol{x}_t)'\boldsymbol{\beta}]^2$$

8 Semi-parametric Estimation

8.1 Maximum Score Estimation (Manski, 1975)

We predict $y_i = 1$ if $\boldsymbol{x}_i'\boldsymbol{\beta} > 0$. We predict $y_i = 0$ if $\boldsymbol{x}_i'\boldsymbol{\beta} \leq 0$. The Score function, which counts the number of correct observations, is defined as

$$\begin{aligned}
S_N(\boldsymbol{\beta}) &:= \sum_{i=1}^{N}\{y_i 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + (1-y_i)1(\boldsymbol{x}_i'\boldsymbol{\beta} \leq 0)\} \\
&= \sum_{i=1}^{N}\{y_i 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + (1-y_i)(1 - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0))\} \\
&= \sum_{i=1}^{N}\{y_i 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + 1 - y_i - (1-y_i)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)\} \\
&= \sum_{i=1}^{N}\{y_i 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + 1 - y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + y_i 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)\} \\
&= \sum_{i=1}^{N}\{(2y_i - 1)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + 1 - y_i\} \\
&= \sum_{i=1}^{N}(2y_i - 1)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + N - \sum_{i=1}^{N} y_i
\end{aligned}$$

The Maximum Score Estimator is

$$\widehat{\boldsymbol{\beta}} = arg\ max_{\boldsymbol{\beta}} \sum_{i=1}^{N}(2y_i - 1)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) \qquad \text{can not use differentiation}$$

MSE can be regarded as Least Absolute Deviation (LAD) Estimator. It can be seen

$$\begin{aligned}
Q_N(\boldsymbol{\beta}) &= N - S_N(\boldsymbol{\beta}) \\
&= \sum_{i=1}^{N}(1 - \{(2y_i - 1)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) + 1 - y_i\}) \\
&= \sum_{i=1}^{N}\{y_i - (2y_i - 1)1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)\} \\
&= \sum_{i=1}^{N}\begin{cases} 1 - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) & \text{if } y_i = 1 \\ 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) & \text{if } y_i = 0 \end{cases} \\
&= \sum_{i=1}^{N}\begin{cases} y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) & \text{if } y_i = 1 \\ -(y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)) & \text{if } y_i = 0 \end{cases} \\
&= \sum_{i=1}^{N}\begin{cases} |y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)| & \text{if } y_i = 1 \text{ as } 1 - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) \geq 0 \\ |y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)| & \text{if } y_i = 0 \text{ as } 0 - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0) \leq 0 \end{cases} \\
&= \sum_{i=1}^{N}|y_i - 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)| \\
&= \sum_{i=1}^{N}|y_i - Median(y_i|\boldsymbol{x}_i)|
\end{aligned}$$

The last line since

$$
\begin{aligned}
Median(y_i|\boldsymbol{x}_i) &= Median(1(y_i^* > 0)|\boldsymbol{x}_i) \\
&= 1(Median(y_i^*|\boldsymbol{x}_i) > 0) \\
&= 1(Median(\boldsymbol{x}_i'\boldsymbol{\beta} + u_i|\boldsymbol{x}_i) > 0) \\
&= 1(Median(\boldsymbol{x}_i'\boldsymbol{\beta}|\boldsymbol{x}_i) + \underbrace{Median(u_i|\boldsymbol{x}_i)}_{0} > 0) && \text{assume 0} \\
&= 1(\boldsymbol{x}_i'\boldsymbol{\beta} > 0)
\end{aligned}
$$

Assume $Median(u_i|\boldsymbol{x}_i) = 0$, $\widehat{\boldsymbol{\beta}}$ is consistent but $N^{1/3}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ converges in distribution to non-normal distribution.

8.2 Semi-parametric MLE (Klein & Spady, 1993)

8.2.1 Algorithm

For $t = 1 \cdots$

Use $\boldsymbol{\beta}^{(t)}$, estimate $F^{(t)}$ by local constant (Nadaraya–Watson) estimator,

$$F^{(t)}(\boldsymbol{x}'\boldsymbol{\beta}^{(t)}) = \frac{\sum_{j=1}^N y_j K((\boldsymbol{x}_j - \boldsymbol{x})'\boldsymbol{\beta}^{(t)}/h)}{\sum_{j=1}^N K((\boldsymbol{x}_j - \boldsymbol{x})'\boldsymbol{\beta}^{(t)}/h)}$$

Given $F^{(t)}$ and $\boldsymbol{\beta}^{(t)}$, compute $\boldsymbol{\beta}^{(t+1)}$ by gradient descent method i.e., $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \alpha \frac{\partial Q_N^{(t)}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}^{(t)}}$ where

$$\frac{\partial Q_N^{(t)}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}^{(t)}} = \sum_{i=1}^N \frac{y_i - F^{(t)}(\boldsymbol{x}_i'\boldsymbol{\beta}^{(t)})}{F^{(t)}(\boldsymbol{x}_i'\boldsymbol{\beta}^{(t)})(1 - F^{(t)}(\boldsymbol{x}_i'\boldsymbol{\beta}^{(t)}))} F^{(t)\prime}(\boldsymbol{x}_i'\boldsymbol{\beta}^{(t)})\boldsymbol{x}_i$$

Loop until convergence of $\boldsymbol{\beta}^{(t)}$.

The advantage of this approach is that we do not have to specify $F(.)$. However, the algorithm is slow by its nature. *np* package in R provides a convenient function for performing the above algorithm.

```r
library(np)

# the first beta is normalized to one
summary(npindex(y ~ x1 + x2, method = "kleinspady", gradients = TRUE, data = d))
```

8.2.2 Asymptotic Distribution

Given certain regularity conditions,

$$\sqrt{N}(\widehat{\boldsymbol{\beta}}_{KS} - \boldsymbol{\beta}_0) \to_d N(\boldsymbol{0}, \{\mathbb{E}[\frac{\partial F(\boldsymbol{x}'\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}(\frac{\partial F(\boldsymbol{x}'\boldsymbol{\beta})}{\partial \boldsymbol{\beta}})' \frac{1}{F(\boldsymbol{x}'\boldsymbol{\beta})(1 - F(\boldsymbol{x}'\boldsymbol{\beta}))}]\}^{-1})$$

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10 Appendix A: Newton's Method

10.1 Objective Function

$f(\boldsymbol{\theta})$ is the objective function. Second order Taylor approximation of $f(\boldsymbol{\theta})$ centered at $\boldsymbol{\theta}_0$ is

$$f(\boldsymbol{\theta}) \approx f(\boldsymbol{\theta}_0) + (\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}})'|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)'\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$$

10.2 First-order Condition

$$\begin{aligned}
\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_1} \approx \frac{\partial[f(\boldsymbol{\theta}_0) + (\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}})'|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)'\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)]}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_1} &= \mathbf{0} \\
\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0} + \frac{1}{2}\frac{\partial(\boldsymbol{\theta} - \boldsymbol{\theta}_0)'}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_1}(\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0} + (\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0})')(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0) &= \mathbf{0} \\
\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0} + \frac{1}{2}\cdot 1 \cdot 2\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0) &= \mathbf{0} \\
\boldsymbol{\theta}_1 = \boldsymbol{\theta}_0 - [\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}]^{-1}\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}
\end{aligned}$$

Thus, pure Newton's method is $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - [\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t}]^{-1}\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_t}$. Damped Newton's method includes the step size t i.e., $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_0 - t[\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}]^{-1}\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}$. The step size can be selected by backtracking line search. Without this, Newton's method may not converge.

10.3 Conditions for Optimization

$$\begin{aligned}
f(\boldsymbol{\theta}_1) \approx f(\boldsymbol{\theta}_0) + \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0) &= f(\boldsymbol{\theta}_0) + \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}(-[\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}]^{-1})\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0} \\
f(\boldsymbol{\theta}_1) - f(\boldsymbol{\theta}_0) \approx \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}(-[\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}]^{-1})\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_0}
\end{aligned}$$

If f is a concave function, $\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}$ is negative semi-definite. Thus, $f(\boldsymbol{\theta}_1) - f(\boldsymbol{\theta}_0) \geq 0$ approximately; If f is a convex function, $\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_0}$ is positive semi-definite. Thus, $f(\boldsymbol{\theta}_1) - f(\boldsymbol{\theta}_0) \leq 0$ approximately.

10.4 Apply to Binary Outcome Model

$$\widehat{\boldsymbol{\beta}}^{+} = \widehat{\boldsymbol{\beta}} - \{\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}\}^{-1}\frac{\partial lnL_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}}$$

For $F(-z) = 1 - F(z)$,

$$\begin{aligned}
&= \widehat{\boldsymbol{\beta}} - \{\sum_{i=1}^{N}[\frac{F((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})F''((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) - F'((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}{F((2y_i-1)\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}]\boldsymbol{x}_i\boldsymbol{x}_i\}^{-1}\sum_{i=1}^{N}\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1-F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}(y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i \\
&= \widehat{\boldsymbol{\beta}} - \{\sum_{i=1}^{N}\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) - 1)\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}\sum_{i=1}^{N}(y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i && \text{If } F = \Lambda \text{ i.e., the logit model} \\
&= \widehat{\boldsymbol{\beta}} - \{-\sum_{i=1}^{N}\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}\sum_{i=1}^{N}(y_i - \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i \\
&= \widehat{\boldsymbol{\beta}} - (-\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{p}) && \text{where } (\boldsymbol{W})_{ii} = p_i(1-p_i) \text{ and } p_i = \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) \\
&= \boldsymbol{I}\widehat{\boldsymbol{\beta}} + (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{I}(\boldsymbol{y} - \boldsymbol{p}) \\
&= (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})\widehat{\boldsymbol{\beta}} + (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{W}\boldsymbol{W}^{-1}(\boldsymbol{y} - \boldsymbol{p}) \\
&= (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{W}\underbrace{(\boldsymbol{X}\widehat{\boldsymbol{\beta}} + \boldsymbol{W}^{-1}(\boldsymbol{y} - \boldsymbol{p}))}_{\boldsymbol{z}}
\end{aligned}$$

Thus, $\widehat{\boldsymbol{\beta}}^{+}$ is a Weighted Least Squares estimator i.e.,

$$\widehat{\boldsymbol{\beta}}^{+} = argmin_{\boldsymbol{\beta}}(\boldsymbol{z}-\boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{W}(\boldsymbol{z}-\boldsymbol{X}\boldsymbol{\beta}) = argmin_{\boldsymbol{\beta}}\sum_{i=1}^{N} w_i(z_i-\boldsymbol{x}_i'\boldsymbol{\beta})^2$$

We loop $\widehat{\boldsymbol{\beta}}^{+} = (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{W}\boldsymbol{z}$ until convergence. Note that $\boldsymbol{W}$ and $\boldsymbol{z}$ is a function of $\widehat{\boldsymbol{\beta}}$.
Note also that $(\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{W}(\boldsymbol{X}\widehat{\boldsymbol{\beta}}+\boldsymbol{W}^{-1}(\boldsymbol{y}-\boldsymbol{p})) = ((\boldsymbol{W}\boldsymbol{X})'\boldsymbol{W}^{-1}(\boldsymbol{W}\boldsymbol{X}))^{-1}(\boldsymbol{W}\boldsymbol{X})'\boldsymbol{W}^{-1}((\boldsymbol{W}\boldsymbol{X})\widehat{\boldsymbol{\beta}}+(\boldsymbol{y}-\boldsymbol{p}))$ as $\boldsymbol{W}$ is diagonal with non-zero elements and thus it is symmetric and invertible.

11 Appendix B: Fisher's Scoring

The difference between Fisher's scoring and Newton's method is that the former uses expected Hessian matrix while the latter uses observed Hessian matrix. *glm*() in R uses Fisher's scoring as the default method.

11.1 Binary Outcome Model is a Non-linear Model

$$\begin{aligned}
y_i &= F(\boldsymbol{x}_i'\boldsymbol{\beta}_0) + e_i \\
&\approx F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) + F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i'(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}) + e_i \\
\underbrace{y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) + F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}}_{y_i^*} &\approx \underbrace{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i'}_{x_i^{*\prime}}\boldsymbol{\beta}_0 + e_i
\end{aligned}$$

where $Var(e_i|\boldsymbol{x}_i) = F(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1 - F(\boldsymbol{x}_i'\boldsymbol{\beta}_0))$.

11.2 Fisher's Scoring for Maximum Likelihood Estimation

$$\widehat{\boldsymbol{\beta}}^+ = \widehat{\boldsymbol{\beta}} - \{\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}]\}^{-1}\frac{\partial lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}} = \widehat{\boldsymbol{\beta}} + \{-\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}]\}^{-1}\frac{\partial lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}}$$

Under regularity conditions, $\widehat{\boldsymbol{\beta}}_{mle} \sim_a N(\boldsymbol{\beta}_0, \{-\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]\}^{-1})$. Therefore, $\{-\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}]\}^{-1}$ is a positive semi-definite and $\mathbb{E}[\frac{\partial^2 lnL_N(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}]$ is a negative semi-definite. This guarantees an increase in likelihood for each iteration. Moreover, the asymptotic covariance matrix is a by-product of Fisher's scoring.

11.3 Apply to Binary Outcome Model

$$\begin{aligned}
\widehat{\boldsymbol{\beta}}^+ &= \widehat{\boldsymbol{\beta}} + \{\sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}\sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}(y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))\boldsymbol{x}_i \\
&= \{\sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}\{\sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i(y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})) + \sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}\} \\
&= \{\sum_{i=1}^N \frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})^2}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i\boldsymbol{x}_i'\}^{-1}\sum_{i=1}^N\{\frac{F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i\}\{y_i - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) + F'(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}\} \\
&= \{\sum_{i=1}^N \frac{1}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i^*\boldsymbol{x}_i^{*\prime}\}^{-1}\{\sum_{i=1}^N \frac{1}{F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(1 - F(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}))}\boldsymbol{x}_i^*\boldsymbol{y}_i^*\}
\end{aligned}$$

It loops until the convergence. Thus, Fisher's scoring is an Iteratively Reweighted Least Squares (IRLS) estimation. Expected Hessian matrix is equal to observed Hessian matrix if $F(.)$ is a logistic function, both are $\sum_{i=1}^N \Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}})(\Lambda(\boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}) - 1)\boldsymbol{x}_i\boldsymbol{x}_i'$ (In general, if $F(.)$ is the canonical mean function in Generalized Linear Model, it can be shown that the observed Hessian matrix with the canonical link function does not depend on y_i). Therefore, Fisher's scoring is the same as Newton's method for the logit model.

11.4 BHHH Algorithm

Information matrix equality is

$$\begin{aligned}
-\mathbb{E}[\frac{\partial^2 lnf(y_i|\boldsymbol{x}_i)}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] &= \mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}_0} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] \\
-\mathbb{E}[\frac{\partial^2 \sum_{i=1}^N lnf(y_i|\boldsymbol{x}_i)}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}] &= \sum_{i=1}^N \mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}|_{\boldsymbol{\beta}_0} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial\boldsymbol{\beta}'}|_{\boldsymbol{\beta}_0}]
\end{aligned}$$

Therefore, if the information matrix equality holds and independence is assumed, Fisher's scoring becomes

$$\widehat{\boldsymbol{\beta}}^{+} = \widehat{\boldsymbol{\beta}} + \{\sum_{i=1}^{N} \mathbb{E}[\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}} \cdot \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}|_{\widehat{\boldsymbol{\beta}}}]\}^{-1} \sum_{i=1}^{N} \frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}}$$

Note that $\frac{\partial lnf(y_i|\boldsymbol{x}_i;\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\widehat{\boldsymbol{\beta}}}$ can be re-used inside the inverse. Note also that the original BHHH algorithm does not include the expectation operator.

12 Appendix C: Gauss-Newton Method

12.1 Objective Function

For non-linear model $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{\theta}_0) + \boldsymbol{\varepsilon}$,

$$f(\boldsymbol{\theta}) = (\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))$$

12.2 Approximate Newton's Method

12.2.1 Gradient Vector

$$\begin{aligned}
\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} &= \frac{\partial(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))}{\partial \boldsymbol{\theta}} \\
&= \frac{\partial(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))'}{\partial \boldsymbol{\theta}}(\boldsymbol{W} + \boldsymbol{W}')(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta})) \\
&= -2(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'})'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))
\end{aligned}$$

12.2.2 Hessian Matrix

$$\begin{aligned}
\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} &= -2\frac{\partial(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'})'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))}{\partial \boldsymbol{\theta}'} \\
&= -2[\frac{\partial^2 \boldsymbol{g}(\boldsymbol{\theta})'}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta})) - (\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'})'\boldsymbol{W}\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}] \\
&\approx 2(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'})'\boldsymbol{W}\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}
\end{aligned}$$

The first term is discarded because $\mathbb{E}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta})) \to_p \boldsymbol{0}$.

12.2.3 Apply Newton's Method

$$\begin{aligned}
\boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - [\frac{\partial^2 f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t}]^{-1}\frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_t} \\
&\approx \boldsymbol{\theta}_t - [2(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t}]^{-1}[-2(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))] \\
&= \boldsymbol{\theta}_t + [(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t}]^{-1}[(\frac{\partial \boldsymbol{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}(\boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta}))]
\end{aligned}$$

It can also be written as

$$= \boldsymbol{\theta}_t - [(\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t}]^{-1}[(\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{\theta}'}|_{\boldsymbol{\theta}_t})'\boldsymbol{W}\boldsymbol{u}]$$

where $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{g}(\boldsymbol{\theta})$.

13 Appendix D: Hessian Matrix of Logit and Probit Model is Negative Definite

13.1 Hessian Matrix

$$\frac{\partial^2 lnL(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = \sum_{i=1}^{N} [-(\frac{y_i}{F(\boldsymbol{x}_i'\boldsymbol{\beta})^2} + \frac{1-y_i}{(1-F(\boldsymbol{x}_i'\boldsymbol{\beta}))^2})F'(\boldsymbol{x}_i'\boldsymbol{\beta})^2 + (\frac{y_i}{F(\boldsymbol{x}_i'\boldsymbol{\beta})} - \frac{1-y_i}{1-F(\boldsymbol{x}_i'\boldsymbol{\beta})})F''(\boldsymbol{x}_i'\boldsymbol{\beta})]\boldsymbol{x}_i\boldsymbol{x}_i'$$

For $y_i = 1$, the summand is $[-\frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta})^2}F'(\boldsymbol{x}_i'\boldsymbol{\beta})^2 + \frac{1}{F(\boldsymbol{x}_i'\boldsymbol{\beta})}F''(\boldsymbol{x}_i'\boldsymbol{\beta})]\boldsymbol{x}_i\boldsymbol{x}_i$

For $y_i = 0$ and $F(-z) = 1 - F(z)$, the summand is $[-\frac{1}{F(-\boldsymbol{x}_i'\boldsymbol{\beta})^2}F'(-\boldsymbol{x}_i'\boldsymbol{\beta})^2 + \frac{1}{F(-\boldsymbol{x}_i'\boldsymbol{\beta})}F''(-\boldsymbol{x}_i'\boldsymbol{\beta})]\boldsymbol{x}_i\boldsymbol{x}_i'$

Thus, it can be written as

$$\begin{aligned}\frac{\partial^2 lnL(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} &= \sum_{i=1}^{N} [-\frac{1}{F((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})^2}F'((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})^2 + \frac{1}{F((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})}F''((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})]\boldsymbol{x}_i\boldsymbol{x}_i \\ &= \sum_{i=1}^{N} [\frac{F((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})F''((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta}) - F'((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})^2}{F((2y_i-1)\boldsymbol{x}_i'\boldsymbol{\beta})^2}]\boldsymbol{x}_i\boldsymbol{x}_i'\end{aligned}$$

If F is log-concave, then $FF'' - F'^2 \leq 0$ and thus the Hessian matrix is a negative semi-definite given $\sum_{i=1}^{N} \boldsymbol{x}_i\boldsymbol{x}_i'$ is a positive definite. As a logistic function and standard normal cdf are log-concave (both standard logistic and standard normal distribution are in the family of log-concave distribution), the Hessian matrix of the logit and the probit model are negative semi-definites. However, if we apply their properties directly, we can show that they are negative definites.

13.2 Properties of Logistic Function

It is easy to verify that $0 < \Lambda(c) < 1$ and $\Lambda(-c) = 1 - \Lambda(c)$ and $\Lambda'(c) = \Lambda(c)(1-\Lambda(c))$ and $\Lambda''(c) = \Lambda(c)(1-\Lambda(c))(1-2\Lambda(c))$ for any c.

13.3 Hessian Matrix of Logit Model

Let $z_i = (2y_i - 1)\boldsymbol{x}_i'\boldsymbol{\beta}$,

$$\begin{aligned}\frac{\partial^2 lnL(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} &= \sum_{i=1}^{N} [\frac{\Lambda(z_i)\Lambda''(z_i) - \Lambda'(z_i)^2}{\Lambda(z_i)^2}]\boldsymbol{x}_i\boldsymbol{x}_i' \\ &= \sum_{i=1}^{N} [\frac{\Lambda(z_i)^2(1-\Lambda(z_i))(1-2\Lambda(z_i)) - \Lambda(z_i)^2(1-\Lambda(z_i))^2}{\Lambda(z_i)^2}]\boldsymbol{x}_i\boldsymbol{x}_i' \\ &= \sum_{i=1}^{N} [1 - 3\Lambda(z_i) + 2\Lambda(z_i)^2 - (1 - 2\Lambda(z_i) + \Lambda(z_i)^2)]\boldsymbol{x}_i\boldsymbol{x}_i' \\ &= \sum_{i=1}^{N} [\Lambda(z_i)^2 - \Lambda(z_i)]\boldsymbol{x}_i\boldsymbol{x}_i' \\ &= \sum_{i=1}^{N} [\Lambda(z_i)(\Lambda(z_i) - 1)]\boldsymbol{x}_i\boldsymbol{x}_i'\end{aligned}$$

The Hessian matrix is a negative definite because $\Lambda(z_i)(\Lambda(z_i) - 1) < 0$ for $\forall i$ and $\sum_{i=1}^{N} \boldsymbol{x}_i\boldsymbol{x}_i'$ is assumed to be a positive definite.

13.4 Properties of Standard Normal Distribution Function

It is easy to verify that $0 < \Phi(c) < 1$ and $\Phi(-c) = 1 - \Phi(c)$ and $\Phi'(c) = \phi(c)$ and $\Phi''(c) = \phi'(c) = -c\phi(c)$ for any c. $\frac{\phi(c)}{\Phi(c)}$ is called inverse Mills' ratio, which is a convex function of c, $\frac{\phi(c)}{\Phi(c)} \to -c$ if $c \to -\infty$, $\frac{\phi(c)}{\Phi(c)} \to 0$ if $c \to \infty$ (Newey and McFadden, 1994, p.2125). This implies $\frac{\phi(c)}{\Phi(c)} > -c$ for all c.

13.5 Hessian Matrix of Probit Model

Let $z_i = (2y_i - 1)\boldsymbol{x}_i'\boldsymbol{\beta}$,

$$\begin{aligned}
\frac{\partial^2 lnL(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'} &= \sum_{i=1}^{N}[\frac{\Phi(z_i)\Phi^{''}(z_i) - \Phi'(z_i)^2}{\Phi(z_i)^2}]\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= \sum_{i=1}^{N}[-z_i\frac{\phi(z_i)}{\Phi(z_i)} - (\frac{\phi(z_i)}{\Phi(z_i)})^2]\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= -\sum_{i=1}^{N}(z_i + \frac{\phi(z_i)}{\Phi(z_i)})\frac{\phi(z_i)}{\Phi(z_i)}\boldsymbol{x}_i\boldsymbol{x}_i'
\end{aligned}$$

As $z_i + \frac{\phi(z_i)}{\Phi(z_i)} > 0$ for $\forall i$ (by its property) and $\sum_{i=1}^{N}\boldsymbol{x}_i\boldsymbol{x}_i'$ is assumed to be a positive definite, the Hessian matrix is thus a negative definite.